

Action-Bregman Information: Directional Value, Mist-Crystal Laws, and Decision Geometry

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Abstract

Shannon mutual information quantifies uncertainty reduction, but many modern information systems are evaluated by the utility of actions taken after receiving a signal. This note develops *Action-Bregman Information* (ABI), a Jensen-gap information functional generated by a soft action potential. ABI is nonnegative, obeys Blackwell monotonicity, and reduces in the hard-rational limit to the expected value of information for a decision problem. Earlier sections record a semantic Fisher law, binary and multiclass mist laws, utility-indexed synergy, directed ABI, semantic eigenlenses, posterior-compression identities, action-policy KL geometry, rationality resonance, soft mist-flash bounds, utility-score sufficiency, null-pair dark information, semantic-dark capacity, semantic water-filling, and semantic refraction. Recent extensions study binary prediction with abstention, semantic water-filling, and semantic-dark capacity. The v1.0 extension treats the inverse rationality parameter β itself as an information-theoretic coordinate. An exact rationality-slope identity is derived: $d\mathcal{I}_{u,\beta}/d\beta$ equals β^{-2} times the Jensen gap of action-policy KL concentration. This shows why ABI need not be monotone in rationality. Choice-preserving evidence can have positive finite-rational value while having zero hard-rational value; this produces confidence-shaping *ghost information*. Two high-rationality extinction laws are derived. For finite posterior-margin atoms, ghost value dies as $w_{\min}e^{-2\beta\delta_{\min}}/\beta$. If the posterior margin density touches the decision boundary as $c\delta^\alpha$, ghost value dies polynomially as $c\Gamma(\alpha+1)\eta_D(\alpha+2)/(2^{\alpha+1}\beta^{\alpha+2})$. Experiments verify the slope identity, expose non-monotone rationality spectra in random finite decision channels, and validate the finite-atom and boundary-density tail constants. Applications are proposed in semantic communication, safety triage, privacy auditing, human/model calibration, and spectrum-aware representation learning.

Keywords: semantic information, value of information, Bregman information, Blackwell informativeness, mutual information, decision theory, semantic communication, classification, finite channels, policy geometry, rationality resonance, soft mist-flash law, semantic efficiency.

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1 Scope and guiding blind spot

Classical information theory gives precise laws for storage, compression, channel capacity, and statistical dependence. Its central measure, mutual information, is invariant to the downstream task: one nat about a nuisance coordinate is counted the same as one nat about a safety-critical coordinate. That invariance is a strength for communication engineering, but it becomes a blind spot when systems are judged by action, control, decision quality, or semantic success.

The working thesis of this note is:

$$\boxed{\text{Information has direction in belief space; utility determines which directions matter.}} \quad (1)$$

The goal is not to replace Shannon information, but to construct task-indexed companions that retain convexity and data-processing behavior while exposing action relevance.

Related foundations include Shannon’s entropy and mutual information [1], Blackwell’s comparison of experiments by value for all decision makers [2, 3, 4], Bregman information and Jensen-gap formulations [5, 6], directed information for feedback and memory [7, 8], and contemporary semantic communication surveys, which repeatedly identify the lack of unified objectives and evaluation criteria as a major obstacle [10, 11].

The claims below are working mathematical results and research targets. Priority over the literature has not been established beyond the cited context. Numerical tests are included to search for contradictions and to expose useful conjectures.

2 Action-Bregman Information

Let Y be a hidden state, S a signal, $a \in \mathcal{A}$ an action, and $u(a, y)$ a utility. For a belief $p \in \Delta_Y$, write $u_a \in \mathbb{R}^{|Y|}$ for the utility vector $u_a(y) = u(a, y)$. Let $\pi(a) > 0$ be a reference action prior and let $\beta > 0$ be an inverse rationality parameter.

Definition 1 (Soft action potential).

$$F_{u,\beta}(p) = \frac{1}{\beta} \log \sum_{a \in \mathcal{A}} \pi(a) \exp\{\beta \langle u_a, p \rangle\}. \quad (2)$$

Definition 2 (Action-Bregman Information).

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) = \mathbb{E}_S F_{u,\beta}(P_{Y|S}) - F_{u,\beta}(P_Y).} \quad (3)$$

As $\beta \rightarrow \infty$, the potential converges to the Bayes value function:

$$F_{u,\infty}(p) = \max_a \langle u_a, p \rangle, \quad (4)$$

so that

$$\boxed{\mathcal{I}_{u,\infty}(Y; S) = \mathbb{E}_S \max_a \mathbb{E}[u(a, Y) \mid S] - \max_a \mathbb{E}[u(a, Y)].} \quad (5)$$

This is the expected value of information for the specified decision problem.

2.1 Bregman form and monotonicity

For a convex differentiable F , define

$$\mathcal{I}_F(Y; S) = \mathbb{E}F(P_{Y|S}) - F(P_Y). \quad (6)$$

Let

$$D_F(q, p) = F(q) - F(p) - \nabla F(p) \cdot (q - p) \quad (7)$$

be the Bregman divergence generated by F .

Proposition 1 (Jensen-Bregman identity).

$$\boxed{\mathcal{I}_F(Y; S) = \mathbb{E}_S D_F(P_{Y|S}, P_Y) \geq 0.} \quad (8)$$

Proof. Since $P_Y = \mathbb{E}_S P_{Y|S}$, the linear term in $D_F(P_{Y|S}, P_Y)$ vanishes after averaging. Nonnegativity follows from convexity. $\square$

Proposition 2 (Blackwell monotonicity). *If $Y \rightarrow S \rightarrow T$, then*

$$\boxed{\mathcal{I}_F(Y; T) \leq \mathcal{I}_F(Y; S).} \quad (9)$$

Proof. The posterior $P_{Y|T}$ is a conditional average of $P_{Y|S}$ given T . Jensen's inequality gives $F(P_{Y|T}) \leq \mathbb{E}[F(P_{Y|S}) | T]$. Averaging over T gives the result. $\square$

Proposition 3 (Chain law). *For two signals (S, T) ,*

$$\boxed{\mathcal{I}_F(Y; S, T) = \mathcal{I}_F(Y; S) + \mathbb{E}_S \left[\mathbb{E}_{T|S} F(P_{Y|S,T}) - F(P_{Y|S}) \right].} \quad (10)$$

Thus ABI has an exact chain rule when the second term is evaluated relative to the posterior prior $P_{Y|S}$.

3 Local geometry: semantic Fisher and decision curvature

The first-order displacement of a posterior has no average effect because posteriors average back to the prior. The leading useful-information term is quadratic.

At small β , let

$$\bar{u} = \mathbb{E}_{a \sim \pi} u_a, \quad M_u = \mathbb{E}_{a \sim \pi} [(u_a - \bar{u})(u_a - \bar{u})^\top]. \quad (11)$$

Then

$$F_{u,\beta}(p) = \bar{u}^\top p + \frac{\beta}{2} p^\top M_u p + O(\beta^2), \quad (12)$$

so

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) = \frac{\beta}{2} \mathbb{E}_S \left[(P_{Y|S} - P_Y)^\top M_u (P_{Y|S} - P_Y) \right] + O(\beta^2).} \quad (13)$$

This is the *semantic Fisher law*: weakly rational useful information is posterior covariance projected through the utility Gram matrix.

A sharper local law holds at arbitrary β .

Proposition 4 (Decision-curvature law). *Let*

$$\omega_a(p) = \frac{\pi(a)e^{\beta\langle u_a, p \rangle}}{\sum_b \pi(b)e^{\beta\langle u_b, p \rangle}} \quad (14)$$

be the Gibbs distribution over actions at belief p . Then

$$\boxed{\nabla^2 F_{u,\beta}(p) = \beta \operatorname{Cov}_{a \sim \omega(p)}(u_a).} \quad (15)$$

Therefore, if $P_{Y|S} = p + \epsilon Z_S$ with $\mathbb{E}Z_S = 0$, then

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) = \frac{\epsilon^2 \beta}{2} \mathbb{E}[Z_S^\top \operatorname{Cov}_{a \sim \omega(p)}(u_a) Z_S] + O(\epsilon^3).} \quad (16)$$

The covariance term is largest when multiple actions are competitive. In the binary correct-prediction problem with $p = P(Y = 1)$,

$$F_\beta(p) = \frac{1}{\beta} \log \left(e^{\beta p} + e^{\beta(1-p)} \right), \quad (17)$$

so

$$\boxed{F''_\beta(p) = \beta \operatorname{sech}^2(\beta(p - 1/2)).} \quad (18)$$

As β grows, useful-information curvature concentrates near the decision boundary $p = 1/2$.

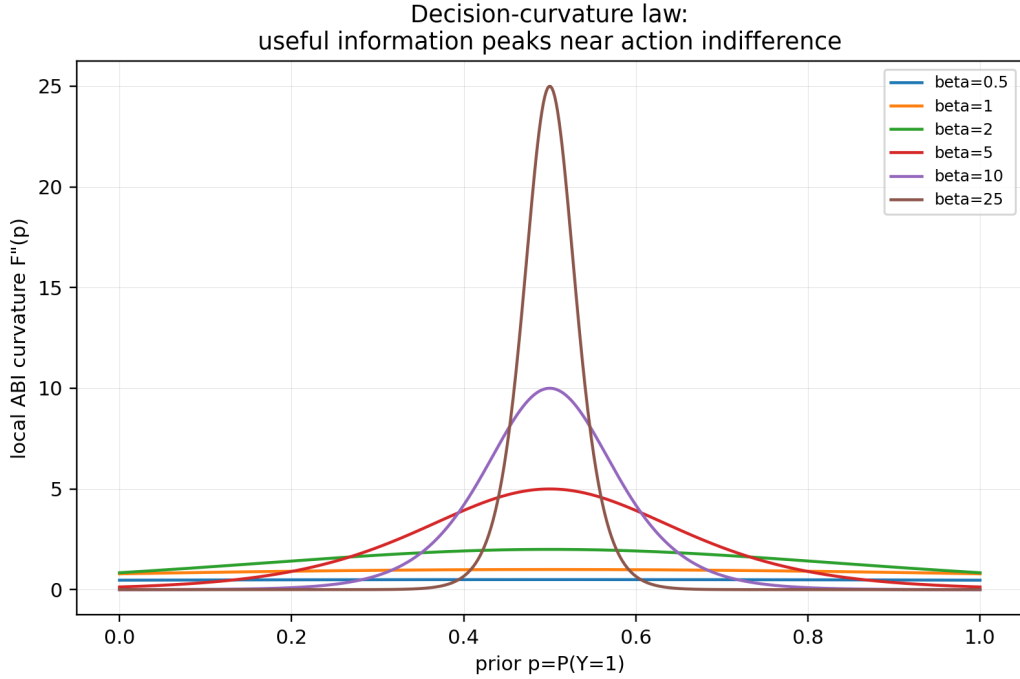


Figure 1: Local ABI curvature for binary prediction. Small posterior perturbations have high value near action indifference and low value when the optimal action is already clear.

4 Binary value-information envelope: the mist-flash law

For $Y \in \{0, 1\}$ with $P(Y = 1) = 1/2$ and hard correct-prediction utility, set

$$q_s = P(Y = 1 \mid S = s), \quad T_s = \left| q_s - \frac{1}{2} \right|. \quad (19)$$

The hard decision value is

$$V(S) = \mathbb{E}T_s, \quad (20)$$

and mutual information in nats is

$$I(Y; S) = \mathbb{E}\phi(T_s), \quad (21)$$

where

$$\phi(t) = \left(\frac{1}{2} + t \right) \log(1 + 2t) + \left(\frac{1}{2} - t \right) \log(1 - 2t). \quad (22)$$

The exact envelope is

$$\boxed{\frac{I(Y; S)}{2 \log 2} \leq V(S) \leq \phi^{-1}(I(Y; S))}. \quad (23)$$

The upper envelope is achieved by weak ubiquitous evidence. The lower envelope is achieved by rare perfect revelations. For small I ,

$$V_{\max}(I) = \sqrt{\frac{I}{2}} - \frac{\sqrt{2}}{12} I^{3/2} + O(I^{5/2}), \quad V_{\min}(I) = \frac{I}{2 \log 2}. \quad (24)$$

Thus two binary signals with equal Shannon information can differ in decision value by a factor of order $I^{-1/2}$ at low information.

5 New result: polysemantic mist-crystal law

The binary mist-flash law has an unexpectedly rich multiclass generalization. Let

$$Y \sim \text{Unif}\{1, \dots, K\}, \quad q_s(i) = P(Y = i \mid S = s), \quad (25)$$

and define top-1 value

$$\boxed{V_K(S) = \mathbb{E}_S \max_i q_s(i) - \frac{1}{K}}. \quad (26)$$

The mutual information is

$$I(Y; S) = \mathbb{E}_S D_{\text{KL}}(q_s \| u_K), \quad u_K = (1/K, \dots, 1/K). \quad (27)$$

5.1 Upper envelope: diffuse mist

For $0 \leq v \leq (K-1)/K$, define

$$\boxed{\Psi_K(v) = \left(\frac{1}{K} + v \right) \log(1 + Kv) + \left(\frac{K-1}{K} - v \right) \log \left(1 - \frac{K}{K-1} v \right)}. \quad (28)$$

This is the KL divergence of the posterior

$$q^\uparrow(v) = \left(\frac{1}{K} + v, \frac{1}{K} - \frac{v}{K-1}, \dots, \frac{1}{K} - \frac{v}{K-1} \right). \quad (29)$$

It is the least informative posterior with top coordinate $1/K + v$, because entropy is maximized by spreading the remaining probability equally.

5.2 Lower envelope: support crystals

For $j = 1, \dots, K$, define the support-crystal point

$$\boxed{I_j = \log \frac{K}{j}, \quad V_j = \frac{1}{j} - \frac{1}{K}.} \quad (30)$$

This point is achieved when the posterior is uniform over a subset of exactly j surviving labels. The signal rules out $K - j$ labels but does not distinguish among the survivors.

Define $L_K(I)$ as the piecewise-linear interpolation through the points (I_j, V_j) . Equivalently, for

$$I \in \left[\log \frac{K}{j+1}, \log \frac{K}{j} \right], \quad j = 1, \dots, K-1, \quad (31)$$

set

$$\boxed{L_K(I) = \frac{1}{j+1} - \frac{1}{K} + \frac{I - \log(K/(j+1))}{j(j+1) \log(1 + 1/j)}.} \quad (32)$$

The slope denominator is the information-value slope between adjacent support crystals:

$$\frac{I_j - I_{j+1}}{V_j - V_{j+1}} = j(j+1) \log \left(1 + \frac{1}{j} \right). \quad (33)$$

Theorem 1 (Polysemantic mist-crystal envelope). *For every finite signal S and uniform K -class target,*

$$\boxed{L_K(I(Y; S)) \leq V_K(S) \leq \Psi_K^{-1}(I(Y; S)).} \quad (34)$$

Both envelopes are attainable by explicit finite experiments.

Proof sketch. For the upper bound, fix $m = \max_i q_i = 1/K + v$. Among all posteriors with this maximum coordinate, entropy is maximized when the remaining mass is spread equally over the other $K - 1$ coordinates. Hence $D_{\text{KL}}(q \| u_K) \geq \Psi_K(v)$. Since Ψ_K is increasing and convex,

$$I = \mathbb{E} D_{\text{KL}}(q_S \| u_K) \geq \mathbb{E} \Psi_K(v_S) \geq \Psi_K(\mathbb{E} v_S) = \Psi_K(V_K). \quad (35)$$

The upper envelope is achieved by a symmetric K -output experiment whose posterior after output s is $q^\uparrow(v)$ centered on label s .

For the lower bound, the entropy-minimizing posterior at fixed maximum probability packs as many coordinates as possible at the maximum value, with at most one fractional remainder and the rest zero. The supporting lines of this maximum-KL curve pass through the uniform-on- j support points (I_j, V_j) ; their slopes decrease with j , so their upper concave envelope is the crystal ladder. Hence $I \leq U_K(V_K)$, where $U_K = L_K^{-1}$ is the piecewise-linear curve through (V_j, I_j) , giving $V_K \geq L_K(I)$. Each crystal point is achieved by a signal that reveals a uniformly random subset of size j containing the true label. Mixtures of adjacent subset experiments achieve the line segments. $\square$

5.3 Low-information asymptotics

At small I ,

$$\boxed{V_{\max, K}(I) = \frac{\sqrt{2(K-1)I}}{K} + \frac{K-2}{3K}I + O(I^{3/2}).} \quad (36)$$

The lower envelope is exactly linear until the first crystal:

$$V_{\min,K}(I) = \frac{I}{K(K-1)\log(K/(K-1))}, \quad 0 \leq I \leq \log \frac{K}{K-1}. \quad (37)$$

Therefore the low-rate value ambiguity obeys

$$\frac{V_{\max,K}(I)}{V_{\min,K}(I)} \sim (K-1) \log \frac{K}{K-1} \sqrt{\frac{2(K-1)}{I}}. \quad (38)$$

For large K , this ratio behaves like $\sqrt{2K/I}$.

Remark 1 (Exceptional multiclass effect). *For $K = 2$, the least actionable low-rate information is a rare perfect revelation. For $K > 2$, this is no longer true. The least actionable low-rate information is a rare class-exclusion event: the signal rules out one label, producing a posterior uniform over $K - 1$ labels. This event carries $\log(K/(K - 1))$ nats but increases top-1 value by only $1/[K(K - 1)]$.*

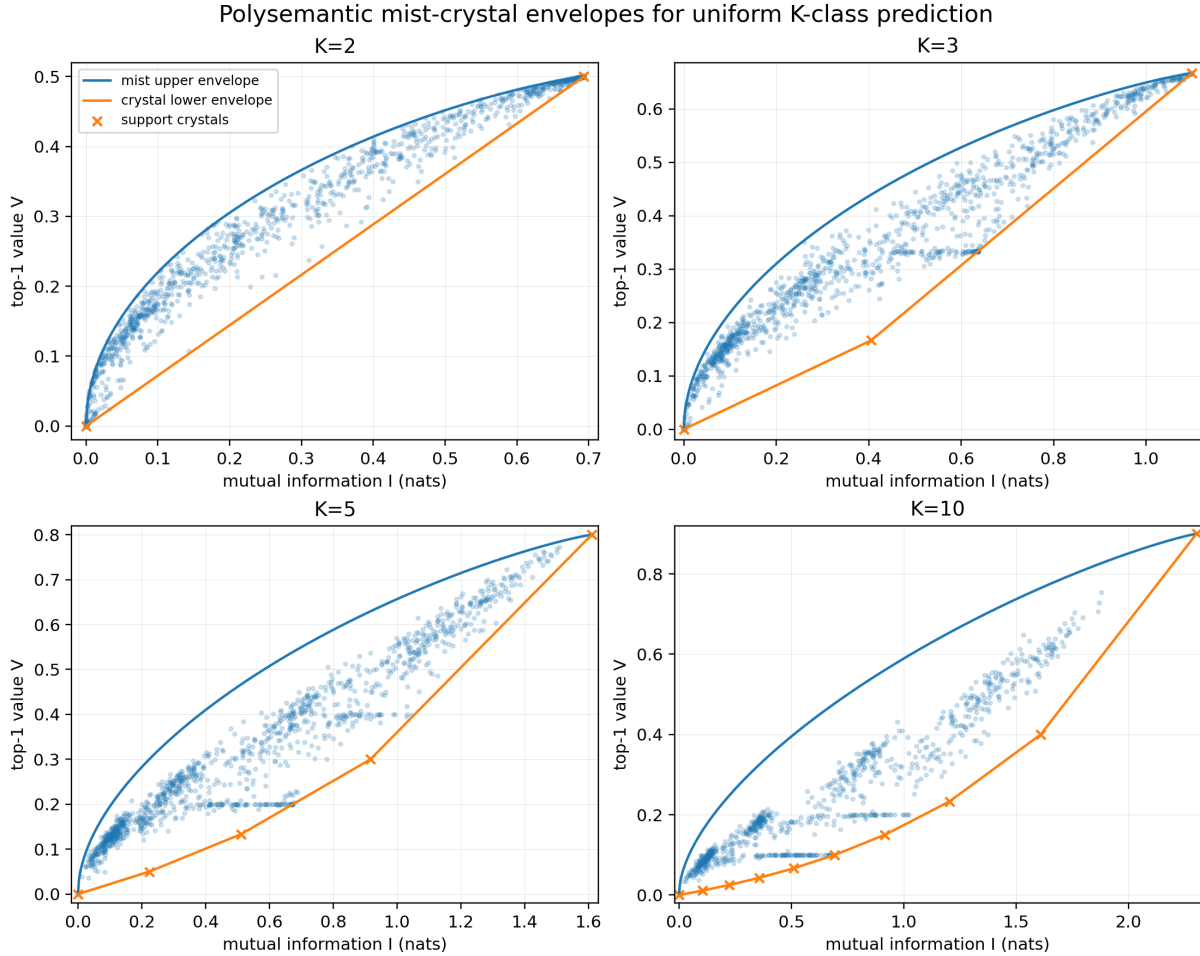


Figure 2: Exact value-information envelopes for uniform K -class prediction. Random finite channels fall between the diffuse mist upper envelope and the support-crystal lower envelope. The lower envelope becomes a ladder for $K > 2$.

6 Numerical experiments

6.1 Random-channel stress test

Finite channels were generated by sampling each row of $P(S|Y)$ from a Dirichlet distribution. For each channel, posteriors were computed exactly from Bayes' rule under the uniform class prior. The experiment varied $K \in \{2, 3, 5, 10, 20\}$, output alphabet sizes 2 to 55, and Dirichlet concentration parameters 0.05 to 5.

Table 1: Envelope stress test over 34,020 random channels. Positive residuals would indicate an envelope violation. Values shown are worst residuals over the sample; observed positives are numerical endpoint roundoff.

K	samples	$\max(V - V_{\max})$	$\max(V_{\min} - V)$
2	7,560	1.55×10^{-15}	0
3	7,560	1.26×10^{-18}	-4.08×10^{-19}
5	7,560	-2.99×10^{-4}	-7.02×10^{-9}
10	7,560	-2.47×10^{-2}	-6.43×10^{-4}
20	3,780	-2.23×10^{-2}	-2.29×10^{-3}

6.2 How large can the value ambiguity be?

The envelope width

$$W_K(I) = \Psi_K^{-1}(I) - L_K(I) \quad (39)$$

measures how much two signals with the same Shannon information can differ in top-1 action value. The width is zero at $I = 0$ and $I = \log K$, and it can be large in the middle.

Table 2: Small-information value ratios. At fixed tiny mutual information, diffuse mist can be far more actionable than crystal-like exclusion information.

K	I	$V_{\min}$	$V_{\max}$	ratio
2	0.001	0.000721	0.022357	30.99
3	0.001	0.000411	0.021189	51.55
5	0.001	0.000224	0.018085	80.71
10	0.001	0.000105	0.013678	129.70
20	0.001	0.000051	0.010041	195.71
2	0.010	0.007213	0.070593	9.79
3	0.010	0.004111	0.067660	16.46
5	0.010	0.002241	0.058444	26.08
10	0.010	0.001055	0.044950	42.62
20	0.010	0.000513	0.033648	65.59

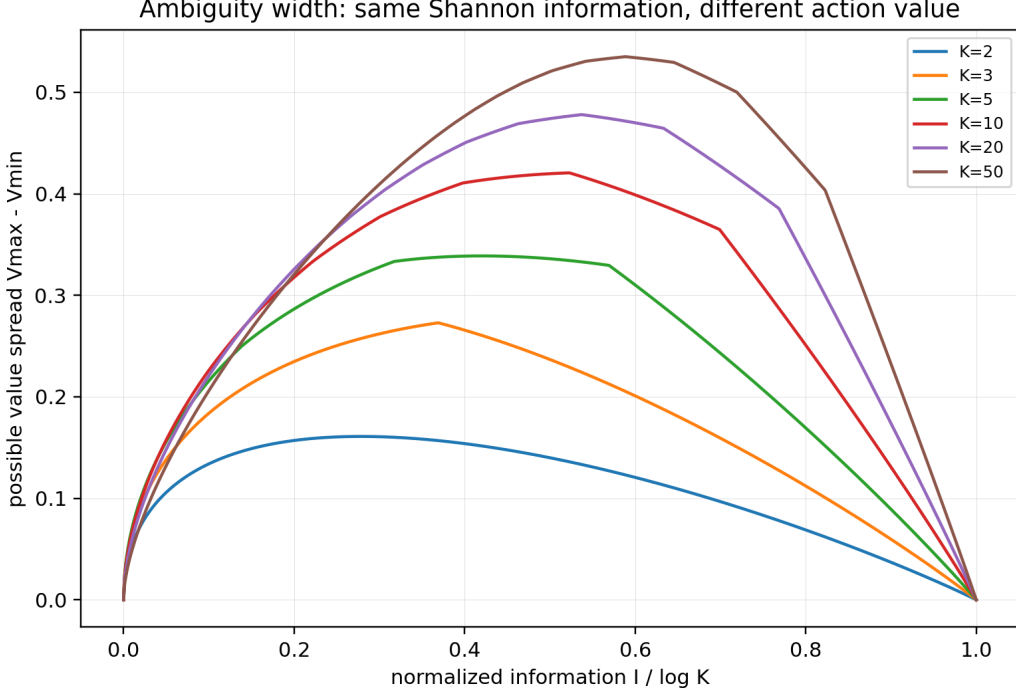


Figure 3: Same mutual information can permit a wide range of top-1 values. The maximum value ambiguity grows with the number of classes over the tested range.

7 Utility-indexed synergy and directed ABI

For two sources X and Z , define the utility-indexed interaction

$$\Delta_{u,\beta}(X, Z \rightarrow Y) = \mathcal{I}_{u,\beta}(Y; (X, Z)) - \mathcal{I}_{u,\beta}(Y; X) - \mathcal{I}_{u,\beta}(Y; Z). \quad (40)$$

Positive values indicate synergy for the decision; negative values indicate redundancy for the decision. This does not attempt to define universal partial-information atoms. It defines decision-relative atoms, which may be better aligned with semantic communication and representation learning.

For a time-indexed process, define directed ABI by

$$\mathcal{I}_{u,\beta}^{\rightarrow}(X^n \rightarrow Y^n) = \sum_{t=1}^n [\mathbb{E}F_{u_t,\beta}(P_{Y_t|Y^{t-1},X^t}) - \mathbb{E}F_{u_t,\beta}(P_{Y_t|Y^{t-1}})]. \quad (41)$$

With log-score geometry this reduces to directed information. With task utility it measures directed useful influence.

8 Local exchange rates: the semantic eigenlens

The mist-crystal envelope shows that equal amounts of Shannon information can have unequal decision value. A local version of the same phenomenon is sharper: near a fixed prior, every posterior direction has a distinct exchange rate from nats to action value.

Let F be twice continuously differentiable near a full-support prior $p \in \Delta_K$. Consider a small finite channel whose posterior has the form

$$P_{Y|S=s} = p + \epsilon Z_s + O(\epsilon^2), \quad \mathbb{E}Z_S = 0, \quad \mathbf{1}^\top Z_s = 0. \quad (42)$$

Let $D_p = \text{diag}(p)$ and set $X_s = D_p^{-1/2} Z_s$. The Shannon mutual information has the standard local form

$$I(Y; S) = \frac{\epsilon^2}{2} \mathbb{E} \|X_S\|^2 + O(\epsilon^3), \quad (43)$$

whereas ABI has the local form

$$\mathcal{I}_F(Y; S) = \frac{\epsilon^2}{2} \mathbb{E} \left[X_S^\top B_F(p) X_S \right] + O(\epsilon^3), \quad (44)$$

with

$$B_F(p) = \Pi_p D_p^{1/2} \nabla^2 F(p) D_p^{1/2} \Pi_p, \quad \Pi_p = I - \sqrt{p} \sqrt{p}^\top. \quad (45)$$

Here $\sqrt{p}$ is the coordinatewise square-root vector; since $\sum_i p_i = 1$, it has unit norm. The projection Π_p removes the direction normal to the probability simplex in the $D_p^{-1/2}$ coordinates.

Proposition 5 (Semantic eigenlens law). *Let $\lambda_1 \leq \dots \leq \lambda_{K-1}$ be the eigenvalues of $B_F(p)$ restricted to the tangent space. For any sufficiently small signal family around p ,*

$$\boxed{\lambda_{\min}^{\text{exc}} + O(\epsilon) \leq \frac{\mathcal{I}_F(Y; S)}{I(Y; S)} \leq \lambda_{\max}^{\text{exc}} + O(\epsilon)}, \quad (46)$$

where the extrema are taken over the posterior directions excited by the channel. In particular, at fixed infinitesimal Shannon information, the maximum useful information is $\lambda_{\max} I + o(I)$ and the minimum is $\lambda_{\min} I + o(I)$.

The matrix $B_F(p)$ is the *semantic eigenlens*. It converts the isotropic local metric of Shannon information into a task-weighted spectrum. Its rank is a local task dimension: the number of posterior directions that carry useful information to second order.

Remark 2 (Semantic-MI inversion). *Suppose two small channels excite eigen-directions with exchange rates $\lambda_h > \lambda_l$. If their Shannon informations satisfy*

$$\frac{I_h}{I_l} > \frac{\lambda_l}{\lambda_h}, \quad (47)$$

then the high-rate direction has larger ABI; but if

$$\boxed{\frac{I_h}{I_l} < \frac{\lambda_l}{\lambda_h}}, \quad (48)$$

then the lower-Shannon-information channel can have higher useful information. If $\lambda_l = 0$, any nonzero signal in a useful direction beats an arbitrary small signal in a purely nuisance direction at the local ABI scale.

8.1 Closed-form spectrum for grouped states

Let $Y = (C, N)$, where $C \in \{1, \dots, R\}$ is the relevant class and $N \in \{1, \dots, G\}$ is a nuisance coordinate. Let the prior be uniform on $K = RG$ states. The action is to predict C , with utility

$$u(a, (c, n)) = \mathbf{1}\{a = c\}. \quad (49)$$

At the uniform prior and with a uniform action reference, the soft action Hessian gives the exact tangent-space spectrum

$$\text{spec}(B_{u,\beta}(p)) = \left\{ \underbrace{\frac{\beta}{R^2}, \dots, \frac{\beta}{R^2}}_{R-1 \text{ times}}, \underbrace{0, \dots, 0}_{K-R \text{ times}} \right\}. \quad (50)$$

Thus the number of actionable local dimensions is $R - 1$, not $RG - 1$. Nuisance directions can carry Shannon information but have zero second-order ABI.

For the tested case $R = 3$, $G = 4$, and $\beta = 6$, the two nonzero exchange rates are exactly $2/3$. Random small-channel tests produced ABI/MI ratios between 0.00321 and 0.42251; the maximum possible local rate $2/3$ was not violated. The median absolute difference between exact ABI/MI and the quadratic eigenlens prediction was 0.00113.

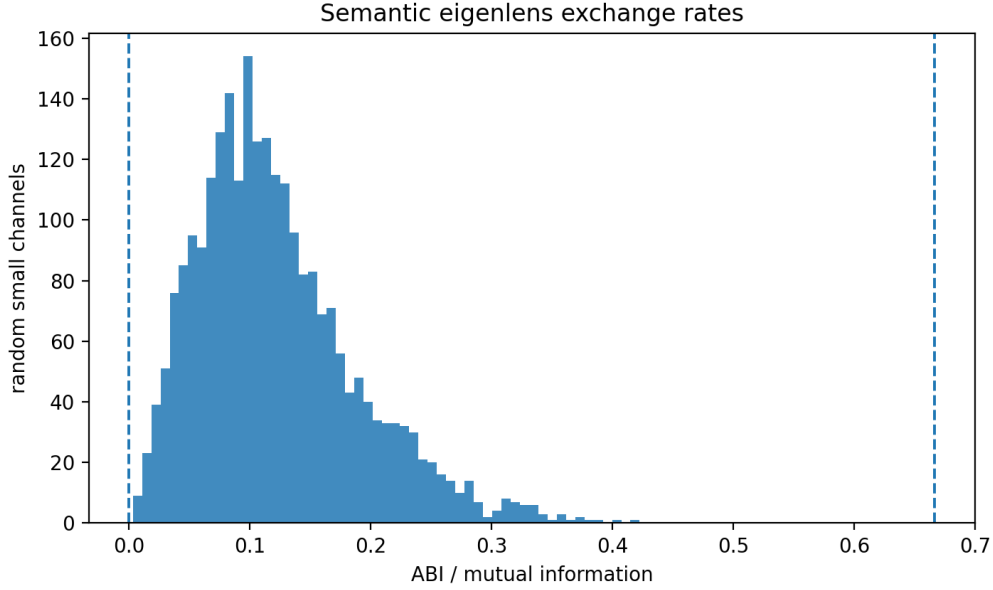


Figure 4: Distribution of exact ABI/MI ratios for random small channels around the uniform prior in a 3×4 grouped-state problem. Dashed lines mark zero and the theoretical maximum local exchange rate $2/3$. Random channels usually mix useful and nuisance directions, so they fall strictly below the maximum.

9 Semantic posterior compression

The eigenlens is local. A global and operational companion is obtained by asking how much ABI is lost when a rich signal alphabet is compressed into fewer messages. Let $q_s = P_{Y|S=s}$ and let $T = g(S)$ be a deterministic compressor. Then

$$q_t = P_{Y|T=t} = \mathbb{E}[q_S \mid T = t]. \quad (51)$$

Proposition 6 (Semantic compression identity). *For any differentiable convex F ,*

$$\boxed{\mathcal{I}_F(Y; S) - \mathcal{I}_F(Y; T) = \mathbb{E}[D_F(q_S, q_T)]}. \quad (52)$$

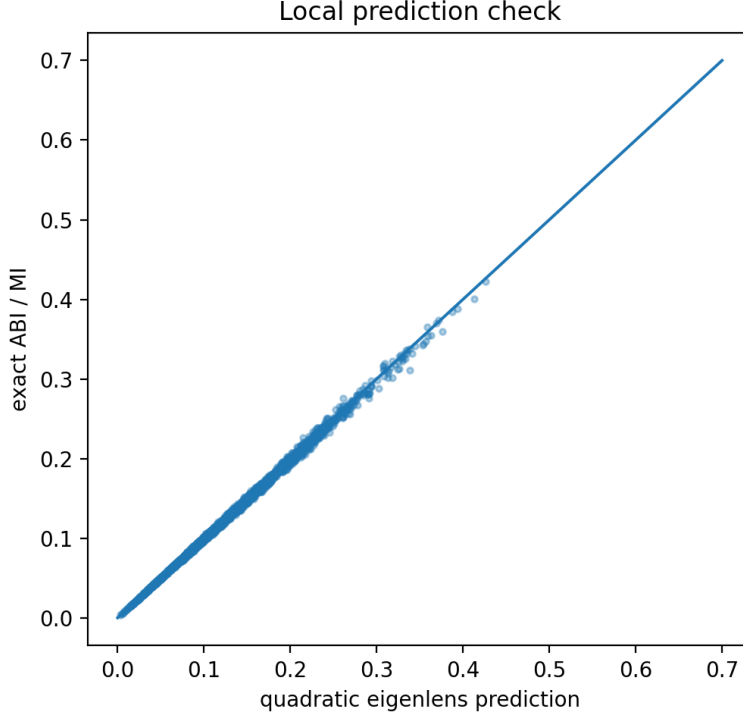


Figure 5: Exact ABI/MI ratios versus the quadratic semantic-eigenlens prediction. The small residuals support the local expansion and expose the usefulness of the eigenlens as a design approximation.

Consequently, among deterministic M -message compressions, preserving ABI is exactly the problem of weighted Bregman quantization of the posterior cloud $\{q_s\}$ under the generator F .

Proof. Expand the right-hand side conditionally on $T = t$:

$$\mathbb{E}[D_F(q_S, q_T) \mid T = t] = \mathbb{E}[F(q_S) \mid T = t] - F(q_t) - \nabla F(q_t)^\top \mathbb{E}[q_S - q_t \mid T = t] \quad (53)$$

$$= \mathbb{E}[F(q_S) \mid T = t] - F(q_t). \quad (54)$$

Averaging over T gives $\mathbb{E}F(q_S) - \mathbb{E}F(q_T)$, which equals $\mathcal{I}_F(Y; S) - \mathcal{I}_F(Y; T)$ because the prior term $F(P_Y)$ cancels. $\square$

This identity is exceptional because it turns semantic compression into geometry. Log-score compression uses $F(q) = \sum_y q_y \log q_y$ and therefore clusters posterior beliefs by KL/Bregman geometry. ABI compression uses the utility-induced $F_{u,\beta}$, so it can deliberately ignore nuisance posterior differences that do not change task value.

9.1 Experiment: posterior compression under nuisance overload

A synthetic posterior cloud was generated over $Y = (R, N)$ with $R = 3$ relevant classes and $N = 4$ nuisance subclasses. The utility was to predict R . The cloud was designed so that many signal symbols strongly moved the nuisance posterior while only some moved the relevant class posterior.

The uncompressed signal had

$$I(Y; S) = 0.84583, \quad \mathcal{I}_{u,8}(Y; S) = 0.19870. \quad (55)$$

The Shannon information decomposed as

$$I(Y; S) = I(R; S) + I(N; S | R) = 0.35249 + 0.49334, \quad (56)$$

so approximately 58.3% of raw mutual information was nuisance information for the task.

Two deterministic compression objectives were compared: semantic Bregman clustering using $F_{u,8}$, and log-score/KL clustering using $F(q) = \sum_y q_y \log q_y$. Both used the same posterior cloud and the same number M of compressed messages.

M	semantic ABI retained	KL ABI retained	semantic MI retained	KL MI retained
2	0.4384	0.3651	0.1391	0.1434
3	0.8425	0.4225	0.2259	0.2125
4	0.9547	0.8012	0.3073	0.3099
6	0.9723	0.8948	0.3320	0.4236
12	0.9869	0.8998	0.3904	0.5447
16	0.9911	0.8883	0.4144	0.5961

Table 3: Semantic and log-score compression of the same posterior cloud. Semantic compression retains far more ABI at very small message budgets, even when it retains less raw mutual information. This is a direct operational form of semantic-MI inversion.

With only $M = 3$ compressed messages, semantic clustering retained 84.25% of ABI, whereas KL clustering retained 42.25%. With $M = 6$, semantic clustering retained 97.23% of ABI while retaining only 33.20% of raw MI; KL clustering retained more raw MI, 42.36%, but less task value, 89.48%. The exact compression identities for ABI and MI held to residuals below 10^{-15} .

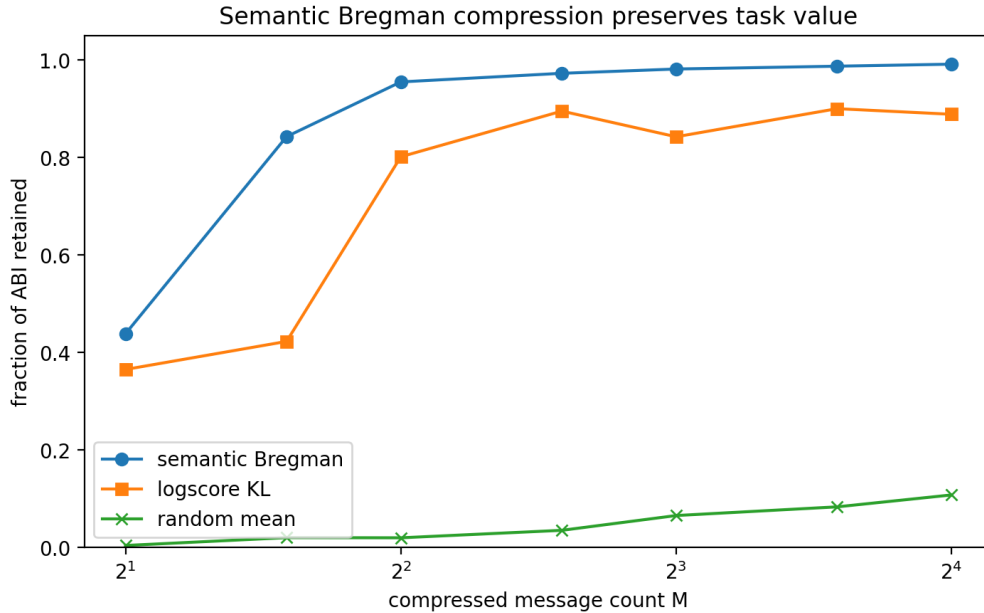


Figure 6: Fraction of ABI retained after compressing the posterior cloud into M messages. Utility-induced Bregman clustering preserves task value much faster than log-score/KL clustering in this nuisance-heavy example.

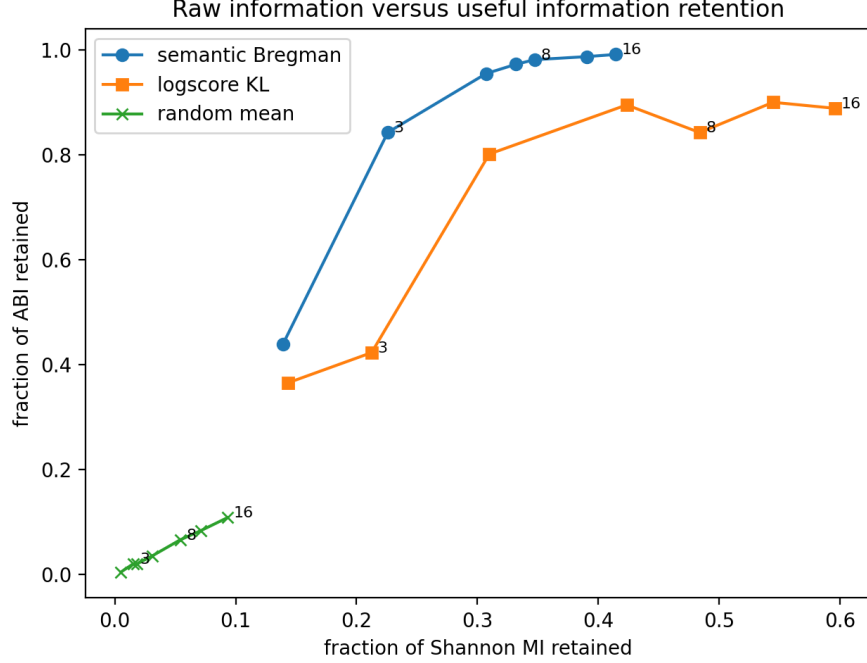


Figure 7: Retention of raw mutual information versus retention of ABI. Semantic compression can move upward without moving far right: it keeps useful information while discarding task-irrelevant information.

Conjecture 1 (Semantic high-resolution law). *For a smooth posterior distribution supported on a d -dimensional manifold in the simplex, the optimal M -message ABI loss should obey a high-resolution quantization law*

$$\inf_{|T| \leq M} (\mathcal{I}_F(Y; S) - \mathcal{I}_F(Y; T)) \sim C_d M^{-2/d} \int \rho(q)^{d/(d+2)} \det(\nabla^2 F(q)|_{T_q})^{1/(d+2)} dq, \quad (57)$$

where ρ is the posterior-cloud density and T_q is the tangent space of the posterior manifold. This would make ABI compression a task-Riemannian quantization problem.

10 Action-policy geometry and rationality resonance

The soft action potential has more structure than a generic convex generator. Because it is a log-partition function over actions, its Bregman geometry can be read directly as a divergence between action policies. This gives a second interpretation of ABI: not only as useful posterior displacement in state-belief space, but as information-induced motion in the decision maker’s own action distribution.

10.1 Gibbs action-policy identity

For a belief p , define the Gibbs soft action policy

$$\omega_p(a) = \frac{\pi(a) \exp\{\beta \langle u_a, p \rangle\}}{\sum_b \pi(b) \exp\{\beta \langle u_b, p \rangle\}}. \quad (58)$$

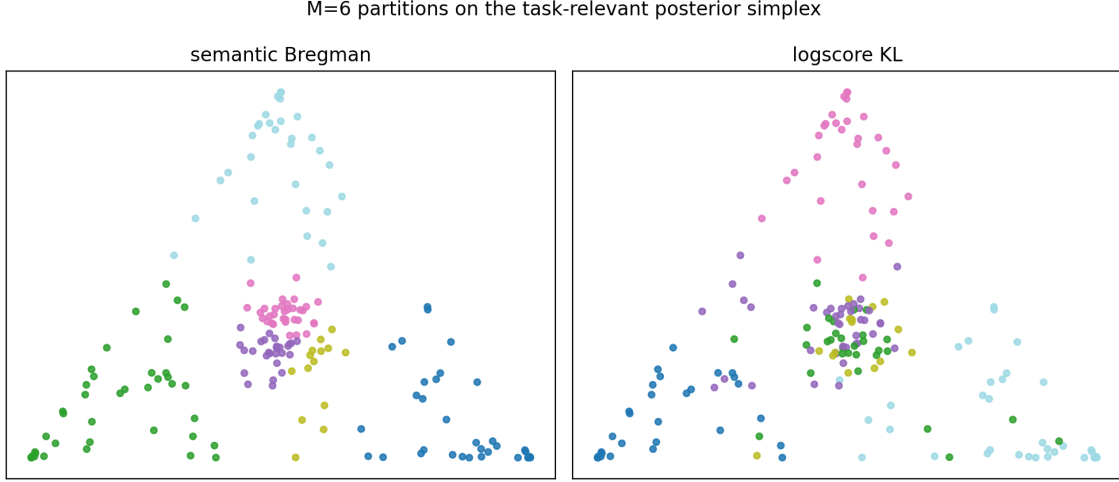


Figure 8: For $M = 6$, semantic Bregman clusters organize the posterior cloud mainly by task-relevant class marginals. KL clusters are pulled more strongly by full-state variation, including nuisance structure. The plot projects posteriors onto the relevant R -simplex.

Theorem 2 (Action-policy KL identity). *For the soft action potential $F_{u,\beta}$,*

$$D_{F_{u,\beta}}(q, p) = \frac{1}{\beta} D_{\text{KL}}(\omega_p \| \omega_q). \quad (59)$$

Therefore

$$\mathcal{I}_{u,\beta}(Y; S) = \frac{1}{\beta} \mathbb{E}_S D_{\text{KL}}(\omega_{P_Y} \| \omega_{P_{Y|S}}). \quad (60)$$

Proof. Let

$$A(\theta) = \log \sum_a \pi(a) e^{\theta \cdot u_a}. \quad (61)$$

Then $F_{u,\beta}(p) = A(\beta p) / \beta$. The Bregman divergence of a log-partition function satisfies

$$A(\theta_q) - A(\theta_p) - \nabla A(\theta_p) \cdot (\theta_q - \theta_p) = D_{\text{KL}}(P_{\theta_p} \| P_{\theta_q}), \quad (62)$$

where P_θ is the exponential-family distribution over actions. Substituting $\theta_p = \beta p$ and $\theta_q = \beta q$ gives the result. The ABI identity follows by averaging $D_F(P_{Y|S}, P_Y)$ over S . $\square$

The orientation is notable. The divergence is not $D_{\text{KL}}(\omega_q \| \omega_p)$, but $D_{\text{KL}}(\omega_p \| \omega_q)$. Thus soft ABI measures how surprising the posterior-adapted action policy would be if judged from the prior soft action policy.

Proposition 7 (Policy loss under deterministic compression). *Let $T = g(S)$ be a deterministic compression and write $q_s = P_{Y|S=s}$ and $q_t = P_{Y|T=t}$. Then*

$$\mathcal{I}_{u,\beta}(Y; S) - \mathcal{I}_{u,\beta}(Y; T) = \frac{1}{\beta} \mathbb{E}_S D_{\text{KL}}(\omega_{q_T} \| \omega_{q_S}), \quad (63)$$

where $q_T = q_{g(S)}$. Thus semantic compression loss is exactly the expected action-policy divergence induced by replacing each detailed posterior with its compressed posterior centroid.

Proof. The posterior-compression identity gives

$$\mathcal{I}_F(Y; S) - \mathcal{I}_F(Y; T) = \mathbb{E}_S D_F(q_S, q_T). \quad (64)$$

Applying the action-policy KL identity to $D_F(q_S, q_T)$ yields the claim. $\square$

Numerical check. The identity was tested on 500 random finite decision problems with random priors, random channels, random utilities, random action priors, and β spanning roughly two orders of magnitude. The maximum absolute residual between ABI and the policy-KL expression was 8.67×10^{-16} ; the median residual was 6.25×10^{-17} .

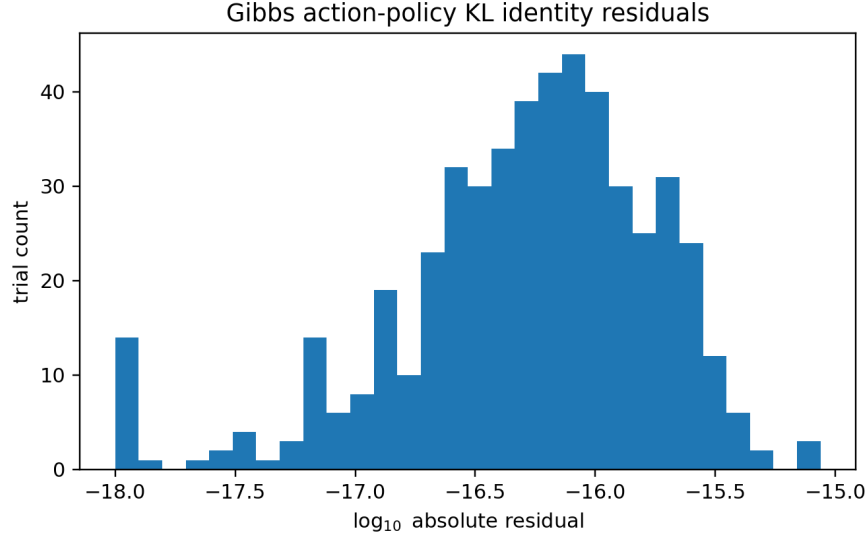


Figure 9: Residuals for the action-policy KL identity over random finite channels and random utilities. The residuals are at numerical roundoff, supporting the exact log-partition derivation.

10.2 Choice-preserving evidence and ghost value

The hard-rational limit counts only information that changes or improves the chosen action. Finite-rational ABI also counts information that sharpens confidence without changing the hard action. This creates a class of signals with zero hard value and positive soft value.

Consider binary correct-prediction utility. Let

$$M_s = P(Y = 1 \mid S = s) - \frac{1}{2}, \quad m = \mathbb{E}M_S = P(Y = 1) - \frac{1}{2}, \quad (65)$$

and define

$$\varphi_\beta(x) = \frac{1}{\beta} \log \cosh(\beta x). \quad (66)$$

Constants cancel in the Jensen gap, giving

$$\boxed{\mathcal{I}_\beta(Y; S) = \mathbb{E}\varphi_\beta(M_S) - \varphi_\beta(m).} \quad (67)$$

Proposition 8 (Choice-preserving soft value). *If $m > 0$ and $M_S > 0$ almost surely, or if $m < 0$ and $M_S < 0$ almost surely, then*

$$\boxed{\mathcal{I}_\infty(Y; S) = 0.} \quad (68)$$

For every finite $\beta > 0$,

$$\boxed{\mathcal{I}_\beta(Y; S) > 0} \quad (69)$$

whenever M_S is not almost surely constant.

Proof. In the hard limit, $\varphi_\infty(x) = |x|$. If all margins have the same sign, then $\mathbb{E}|M_S| = |\mathbb{E}M_S|$, so the hard Jensen gap is zero. For finite β , φ_β is strictly convex because $\varphi_\beta''(x) = \beta \operatorname{sech}^2(\beta x) > 0$. Jensen's inequality is strict unless M_S is constant. $\square$

This finite-rational surplus may be called *ghost value*: it is invisible to the deterministic Bayes action but visible to a soft or confidence-sensitive decision maker.

10.3 Rationality resonance law

Suppose the posterior margin is a small perturbation around a nonzero prior margin,

$$M_S = m + \epsilon Z_S, \quad \mathbb{E}Z_S = 0, \quad m \neq 0. \quad (70)$$

If ϵ is small enough that the sign of the margin is preserved, then the hard value is zero. The soft value has a universal boundary-layer profile.

Theorem 3 (Small-variance rationality resonance). *For fixed $m \neq 0$,*

$$\boxed{\mathcal{I}_\beta(Y; S) = \frac{\epsilon^2}{2} \beta \operatorname{sech}^2(\beta m) \mathbb{E}Z_S^2 + O(\epsilon^3).} \quad (71)$$

The leading term is maximized at

$$\boxed{\beta_* |m| = x_*, \quad x_* \tanh x_* = \frac{1}{2},} \quad (72)$$

with

$$\boxed{x_* = 0.771702319209 \dots} \quad (73)$$

The maximum leading coefficient is

$$\boxed{\mathcal{I}_{\beta_*}(Y; S) = C_* \frac{\epsilon^2 \mathbb{E}Z_S^2}{|m|} + O(\epsilon^3), \quad C_* = \frac{x_*}{2} \operatorname{sech}^2 x_* = 0.223871602347 \dots} \quad (74)$$

Proof. Taylor expansion around m gives

$$\mathbb{E}\varphi_\beta(m + \epsilon Z_S) - \varphi_\beta(m) = \frac{\epsilon^2}{2} \varphi_\beta''(m) \mathbb{E}Z_S^2 + O(\epsilon^3). \quad (75)$$

Since $\varphi_\beta''(m) = \beta \operatorname{sech}^2(\beta m)$, the expansion follows. Let $x = \beta |m|$. Maximizing $x \operatorname{sech}^2 x$ gives

$$\frac{d}{dx} (x \operatorname{sech}^2 x) = \operatorname{sech}^2 x (1 - 2x \tanh x), \quad (76)$$

so the unique positive maximizer satisfies $x \tanh x = 1/2$. $\square$

The resonance law says that useful finite-rational information is largest when the decision temperature $1/\beta$ is comparable to the prior choice margin. If β is too small, the decision maker is too noisy to exploit the evidence. If β is too large, the decision maker has already committed to a hard action and confidence-only evidence disappears.

Two-point experiment. A symmetric choice-preserving signal has margins

$$M_S = m + a \quad \text{or} \quad m - a \quad (77)$$

with equal probability and $0 < a < m$. The exact soft value is

$$G_\beta(m, a) = \frac{1}{2}\varphi_\beta(m + a) + \frac{1}{2}\varphi_\beta(m - a) - \varphi_\beta(m). \quad (78)$$

For $a/m \ll 1$, its peak occurs at $\beta m = x_*$. Numerically, as a/m increases toward the decision boundary, the peak drifts slowly upward rather than breaking the law abruptly.

a/m	peak $\beta_* m$	peak constant mG_{β_*}/a^2
0.05	0.771958	0.223886
0.15	0.774016	0.224001
0.35	0.784564	0.224584
0.75	0.836931	0.227250
0.95	0.885789	0.229435

Table 4: Finite-amplitude drift of the rationality-resonance peak for the two-point choice-preserving signal. The small-amplitude constants are $x_* = 0.771702319\dots$ and $C_* = 0.223871602\dots$

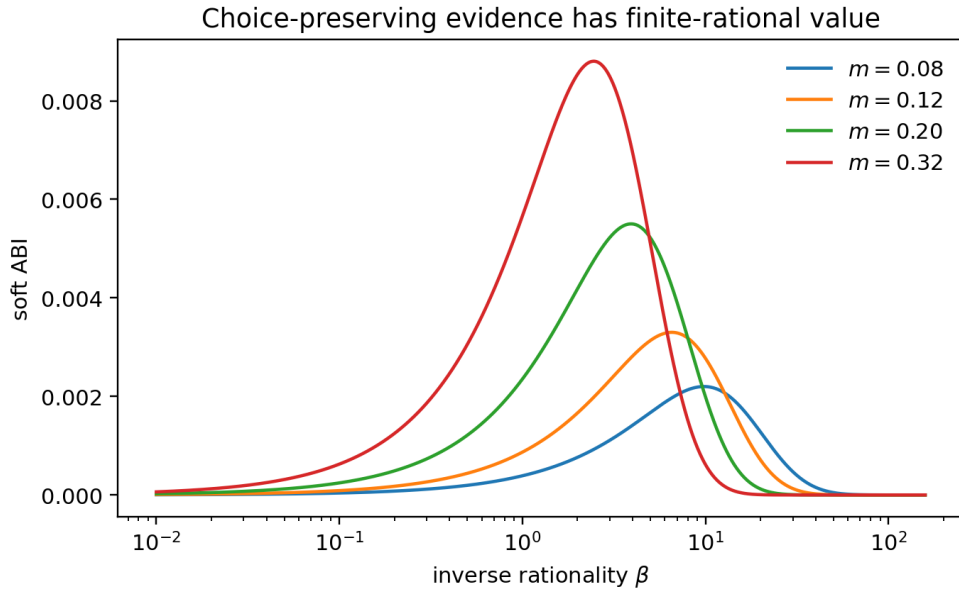


Figure 10: Soft ABI as a function of inverse rationality for choice-preserving binary evidence. The hard value is zero for every curve, yet each finite-rational curve has a positive interior maximum.

10.4 Interpretation and applications of the resonance

The rationality-resonance law separates three kinds of information:

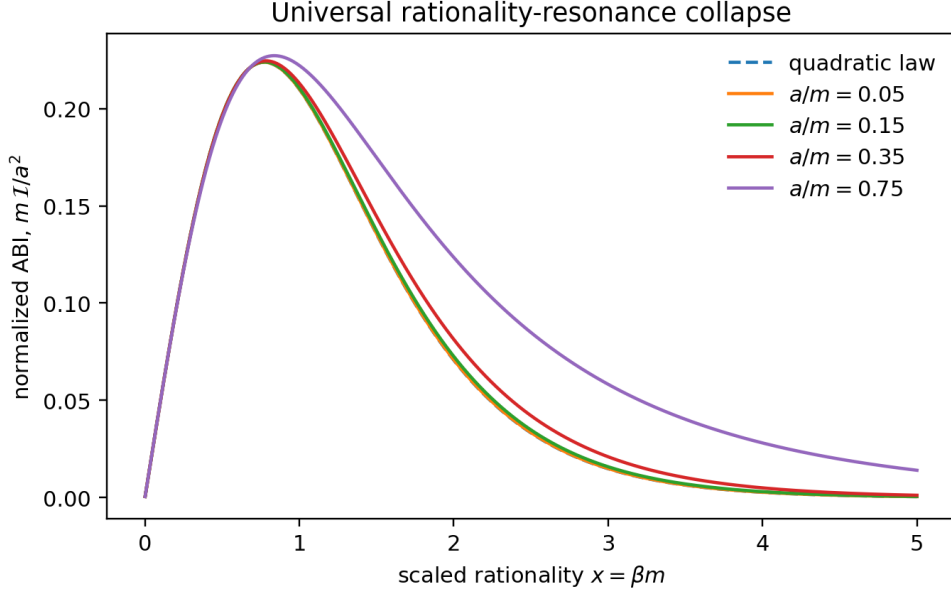


Figure 11: Universal collapse after scaling by the prior margin and posterior-margin amplitude. For small a/m , the exact curve follows $(1/2)x \operatorname{sech}^2 x$ with $x = \beta m$. Larger amplitudes drift gently as the lower posterior margin approaches the choice boundary.

1. *Choice-changing information*, which changes the hard Bayes action and survives as $\beta \rightarrow \infty$.
2. *Confidence-shaping information*, which does not change the hard action but changes soft action probabilities and confidence.
3. *Semantic-null information*, which changes only task-irrelevant posterior directions and has zero ABI even at finite β .

This distinction suggests a design principle for task-aware systems. If the receiver is a calibrated probabilistic decision maker, confidence-shaping evidence has operational value and should not be discarded. If the receiver is a deterministic actuator with no confidence channel, choice-preserving evidence can be compressed away after safety margins are verified. Recent decision-theoretic bottleneck and task-aware semantic communication work also emphasizes that task relevance, not raw mutual information alone, should drive representation and modality selection [12, 13]. The resonance result adds a temperature-dependent rule: the same evidence can be valuable or valueless depending on the receiver’s rationality scale.

Conjecture 2 (Multiaction resonance layers). *For a general finite action set, finite-rational ABI concentrates near the pairwise indifference hyperplanes*

$$\langle u_a - u_b, p \rangle = 0. \quad (79)$$

Small posterior noise at distance d from the nearest active indifference face has a leading response proportional to

$$\beta e^{-c\beta d} \quad (80)$$

for an action-pair-dependent constant c , with a resonance at $\beta d = O(1)$. The binary law gives the exact constant and profile for the two-action case.

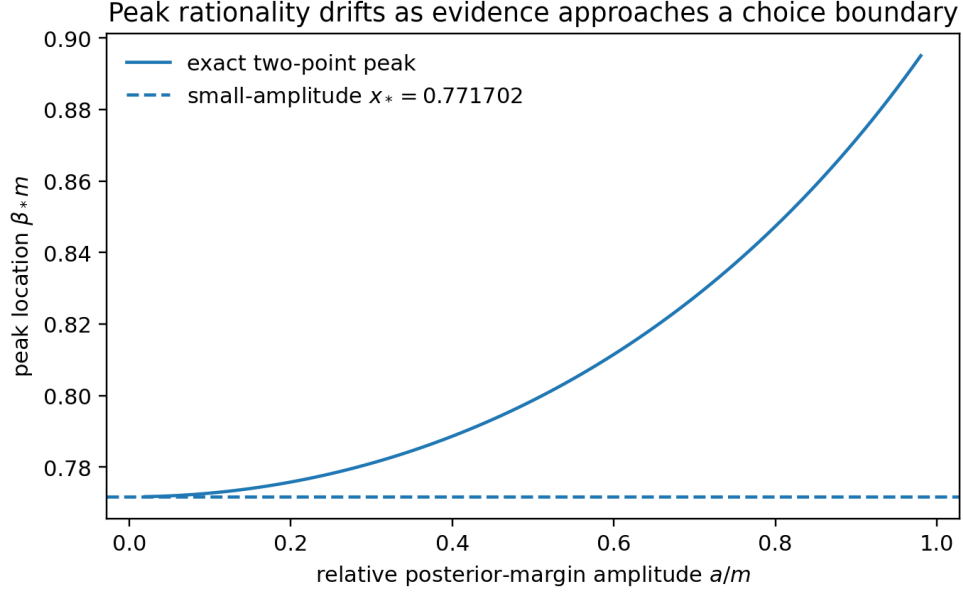


Figure 12: Peak location $\beta_* m$ for two-point choice-preserving evidence as a function of relative amplitude a/m . The dashed line is the universal small-amplitude constant x_* .

11 New result: finite-rational mist–flash law

The hard binary mist–flash law in Section 4 compares top-1 decision value with Shannon mutual information. The soft action-potential geometry allows a sharper finite-rational version. It is exceptional because finite rationality turns the low-information square-root hard-value law into a linear law with an exact slope, while preserving a full global envelope.

Assume throughout this section that $Y \sim \text{Bernoulli}(1/2)$ and that the utility is correct binary prediction. Write

$$q_s = P(Y = 1 \mid S = s), \quad T_s = |q_s - 1/2|. \quad (81)$$

As in the hard law,

$$I(Y; S) = \mathbb{E} \phi(T_S), \quad (82)$$

where

$$\phi(t) = \left(\frac{1}{2} + t\right) \log(1 + 2t) + \left(\frac{1}{2} - t\right) \log(1 - 2t), \quad 0 \leq t \leq \frac{1}{2}. \quad (83)$$

For finite inverse rationality β , the soft ABI contribution of posterior offset t is

$$\chi_\beta(t) = \frac{1}{\beta} \log \cosh(\beta t). \quad (84)$$

Thus

$$\mathcal{I}_\beta(Y; S) = \mathbb{E} \chi_\beta(T_S). \quad (85)$$

Theorem 4 (Soft binary mist–flash envelope). *Let $I = I(Y; S)$ and let $\phi^{-1} : [0, \log 2] \rightarrow [0, 1/2]$ denote the inverse of ϕ . Then*

$$\boxed{\frac{\chi_\beta(1/2)}{\log 2} I \leq \mathcal{I}_\beta(Y; S) \leq \chi_\beta(\phi^{-1}(I))}. \quad (86)$$

The upper curve is achieved by symmetric ubiquitous evidence with $T_S \equiv \phi^{-1}(I)$. The lower line is achieved by a rare-flash experiment that is perfectly revealing with probability $I/\log 2$ and uninformative otherwise.

Proof. Define

$$h_\beta(x) = \chi_\beta(\phi^{-1}(x)), \quad 0 \leq x \leq \log 2. \quad (87)$$

The claim reduces to proving that h_β is concave, increasing, and satisfies $h_\beta(0) = 0$. For $x = \phi(t)$,

$$\frac{d^2 h_\beta}{dx^2} = \frac{\chi''_\beta(t)\phi'(t) - \chi'_\beta(t)\phi''(t)}{\phi'(t)^3}. \quad (88)$$

Now

$$\chi'_\beta(t) = \tanh(\beta t), \quad \chi''_\beta(t) = \beta \operatorname{sech}^2(\beta t), \quad (89)$$

and

$$\phi'(t) = 2 \operatorname{arctanh}(2t), \quad \phi''(t) = \frac{4}{1 - 4t^2}. \quad (90)$$

Therefore $h''_\beta \leq 0$ is equivalent to

$$\frac{2\beta}{\sinh(2\beta t)} \leq \frac{2}{(1 - 4t^2) \operatorname{arctanh}(2t)}, \quad (91)$$

or

$$\beta(1 - 4t^2) \operatorname{arctanh}(2t) \leq \sinh(2\beta t). \quad (92)$$

Let $x = 2t$. The elementary inequality $(1 - x^2) \operatorname{arctanh} x \leq x$ on $0 \leq x < 1$ gives

$$\beta(1 - 4t^2) \operatorname{arctanh}(2t) \leq 2\beta t, \quad (93)$$

while $\sinh(2\beta t) \geq 2\beta t$. Hence h_β is concave.

Since $\mathcal{I}_\beta = \mathbb{E}h_\beta(\phi(T_S))$ and $I = \mathbb{E}\phi(T_S)$, Jensen's inequality gives the upper bound. Since a nonnegative concave function on $[0, L]$ lies above its chord between 0 and L , with $L = \log 2$, one has

$$h_\beta(x) \geq \frac{x}{\log 2} h_\beta(\log 2) = \frac{x}{\log 2} \chi_\beta(1/2), \quad (94)$$

which gives the lower bound after averaging. The stated experiments achieve equality in the two bounds. $\square$

11.1 Semantic efficiency cap

A direct corollary is a global soft semantic efficiency cap:

$$\boxed{\mathcal{I}_\beta(Y; S) \leq \frac{\beta}{4} I(Y; S).} \quad (95)$$

Indeed, h_β is concave and $h'_\beta(0) = \beta/4$. Thus finite-rational binary prediction cannot extract more than $\beta/4$ soft-action nats per Shannon nat. The cap is approached by vanishingly weak ubiquitous evidence.

The small-rate expansion of the optimal upper curve is

$$\boxed{\chi_\beta(\phi^{-1}(I)) = \frac{\beta}{4} I - \frac{\beta(\beta^2 + 4)}{48} I^2 + O(I^3).} \quad (96)$$

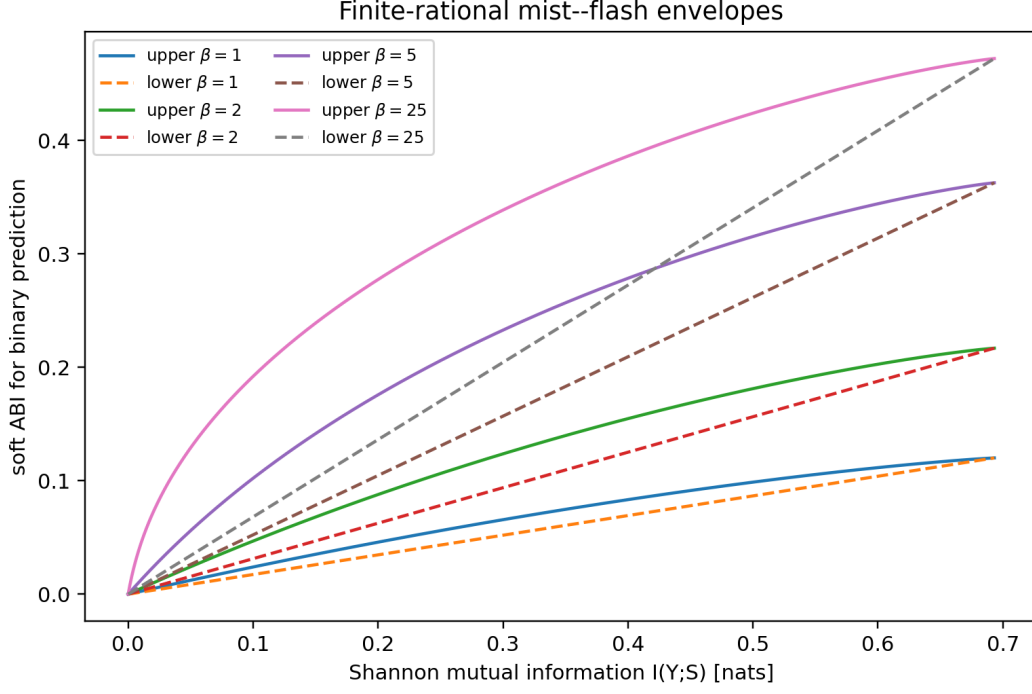


Figure 13: Exact finite-rational mist–flash envelopes for binary prediction. For every fixed β , soft ABI at fixed Shannon mutual information lies between a rare-flash line and a diffuse-mist curve.

The lower flash line has slope

$$\rho_{\beta}^{\text{flash}} = \frac{\log \cosh(\beta/2)}{\beta \log 2}. \quad (97)$$

Consequently, for a finite-rational decision maker, the low-information advantage of mist over flash is bounded by a constant ratio,

$$\frac{\rho_{\beta}^{\text{mist}}}{\rho_{\beta}^{\text{flash}}} = \frac{\beta^2 \log 2}{4 \log \cosh(\beta/2)}, \quad (98)$$

whereas in the hard limit the advantage diverges as $I^{-1/2}$.

11.2 Rationality–information boundary layer

Let

$$U_{\beta}(I) = \chi_{\beta}(\phi^{-1}(I)) \quad (99)$$

be the optimal soft value at Shannon information I . Holding

$$x = \beta \phi^{-1}(I) \quad (100)$$

fixed gives the exact scaling identity

$$\boxed{\beta U_{\beta}(\phi(x/\beta)) = \log \cosh x} \quad (101)$$

whenever $0 \leq x \leq \beta/2$. Thus finite rationality creates a boundary layer at

$$I = O(\beta^{-2}). \quad (102)$$

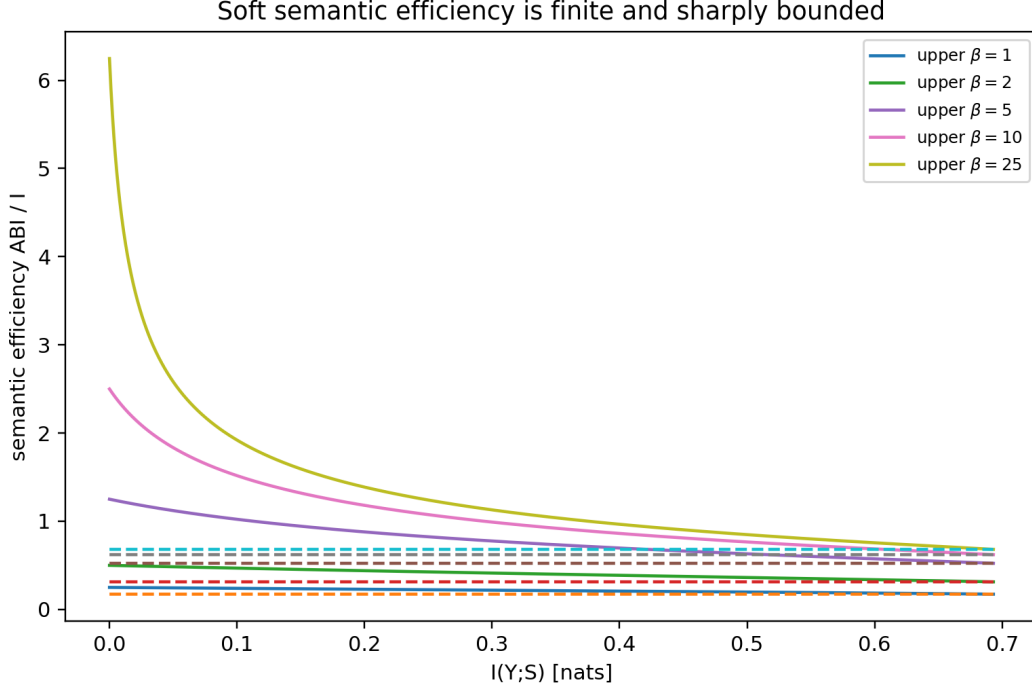


Figure 14: Soft semantic efficiency $\mathcal{I}_\beta/I$ is sharply bounded. The upper efficiency tends to $\beta/4$ at vanishing information and decreases with information; the lower efficiency is the rare-flash slope.

For $x \ll 1$,

$$\log \cosh x = \frac{x^2}{2} + O(x^4), \quad (103)$$

which is the linear-information regime $U_\beta(I) \sim \beta I/4$. For $x \gg 1$,

$$\log \cosh x = x - \log 2 + o(1), \quad (104)$$

which approaches hard decision value up to the finite action-entropy penalty $(\log 2)/\beta$.

This shows that the hard square-root law is not simply the $\beta \rightarrow \infty$ version of the finite- β linear law. The two limits are separated by the scaling window $I \asymp \beta^{-2}$.

11.3 Numerical stress test

Random binary channels were sampled with output alphabet sizes between 2 and 24. For each sampled channel and each value of β , the posterior offsets T_S were computed exactly from Bayes' rule and then compared with the envelope. Across 3500 random channels for each of six values of β , no material violations were found. The largest upper-envelope residual was

$$1.70 \times 10^{-12}, \quad (105)$$

consistent with interpolation error in ϕ^{-1} ; the concavity diagnostic for h_β had no positive numerical residual on the tested grid.

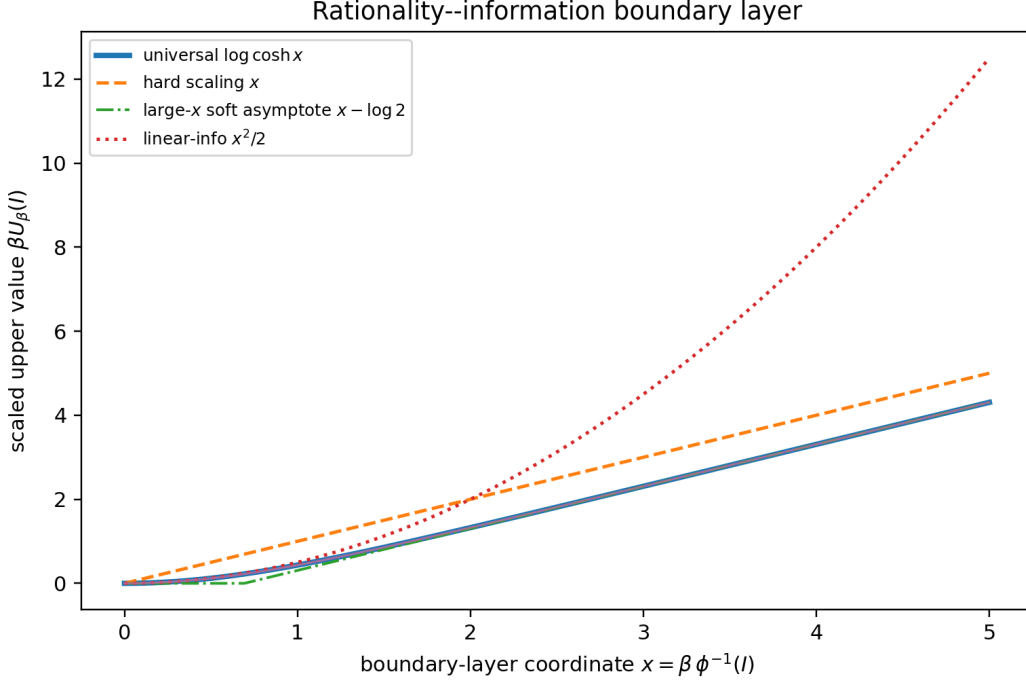


Figure 15: Universal boundary layer between the low-information linear soft regime and the hard decision regime. The scaled optimal value is $\log \cosh x$, where $x = \beta \phi^{-1}(I)$.

11.4 Interpretation

The hard mist–flash law says that at tiny Shannon budgets, diffuse weak evidence can be much more actionable than rare perfect evidence. The soft law refines this: finite rationality regularizes the divergence. A finite- β agent values infinitesimal evidence quadratically in posterior displacement and therefore linearly in mutual information. Superlinear $\sqrt{I}$ extraction appears only when rationality is high enough to resolve evidence whose posterior displacement is of order $1/\beta$.

This suggests a concrete engineering rule. In low-rate semantic channels, increasing decoder rationality and increasing Shannon rate are partially substitutable only through the boundary-layer variable $\beta \phi^{-1}(I) \approx \beta \sqrt{I/2}$. A system operating below this scale is information-limited by soft confidence rather than by hard choice accuracy.

12 Applications

Semantic communication. ABI supplies a task-indexed objective for semantic communication systems: maximize $\mathcal{I}_{u,\beta}$ per bit, joule, second, or channel use. The mist-crystal law shows that two representations with equal mutual information can have very different task value, so a semantic codec should care about posterior direction, not only information magnitude. The compression identity gives a concrete design rule: encode posterior beliefs using the Bregman geometry generated by the downstream utility.

Sensor design. The semantic Fisher and decision-curvature laws turn sensor design into an eigenvalue problem. Sensors should move posteriors along high-curvature utility directions and near

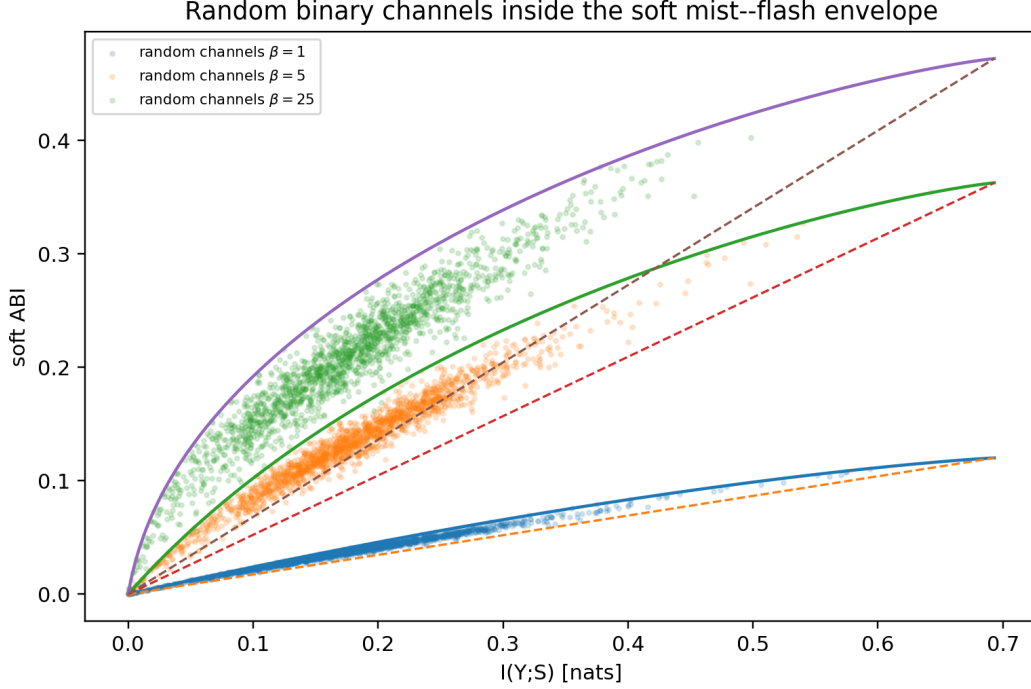


Figure 16: Random binary channels lie inside the finite-rational mist-flash envelopes. The cloud is generated from random output alphabets and Dirichlet-distributed channel rows.

choice boundaries where small evidence can change action quality.

Privacy and safety. Raw mutual information can overstate leakage about nuisance data and understate leakage about low-dimensional but action-critical attributes. ABI gives a utility-indexed privacy risk: how much a signal improves an adversary’s decision value for a specified action class.

Classification diagnostics. The mist-crystal envelope gives exact benchmarks relating mutual information and achievable top-1 value. It can diagnose whether a representation behaves like diffuse evidence, rare exclusion information, or rare identification information. The finite-rational binary envelope adds a calibration for soft classifiers: at a given inverse rationality β , no binary representation can exceed $\beta/4$ soft ABI per Shannon nat at vanishing rate.

Low-rate alerting and querying. The rationality–information boundary layer shows when weak evidence should be accumulated rather than replaced by rare high-certainty alerts. If $\beta\sqrt{I/2} \ll 1$, soft value is approximately linear in I ; if $\beta\sqrt{I/2} = O(1)$, the system crosses into the hard-choice regime. This can guide alarm thresholds, active querying, and short-packet semantic communication.

Causal influence. Directed ABI can separate statistically real causal influence from decision-relevant causal influence. This is useful when a process drives nuisance variation in a target but not the task-critical component.

13 Targets for further study

1. **Nonuniform priors.** Derive exact mist-crystal envelopes when Y is not uniform. The lower envelope should become a prior-dependent face lattice of the probability simplex.
2. **Top- r and abstention decisions.** Replace top-1 utility with top- r success or abstention/rejection utility. The crystal ladder is expected to change its knots.
3. **Semantic channel capacity.** Define

$$C_{u,\beta}(W) = \max_{P_X} \mathcal{I}_{u,\beta}(Y; S) \quad (106)$$

for a channel family or sensing model, then compare it with Shannon capacity and rate-distortion functions.

4. **Finite-blocklength ABI.** Develop nonasymptotic bounds for expected utility improvement under short packets and latency constraints.
5. **Estimator theory.** Develop finite-sample estimators for ABI, directed ABI, and utility-indexed synergy that avoid unstable high-dimensional mutual-information estimation.
6. **Action-information inequalities.** Search for inequalities satisfied by Bregman/Jensen information functionals beyond Shannon-type inequalities.
7. **Representation audits.** Use the value-width $W_K(I)$ and semantic eigenlens spectra to audit whether learned representations are actionable or merely informative.
8. **ABI rate-distortion.** Use the compression identity to define finite-message and rate-constrained semantic quantization objectives based on $D_F(q, c)$ rather than KL distortion.
9. **Semantic eigenbasis learning.** Estimate leading generalized eigenvectors of $(\nabla^2 F(p), \text{diag}(1/p))$ from samples and use them as task-relevant representation axes.
10. **Nullspace audits.** Measure how much mutual information in a representation lies in semantic-null directions for a chosen action potential.
11. **Policy-level ABI estimation.** Use the action-policy KL identity to estimate ABI from observed soft action distributions when posterior beliefs are unavailable.
12. **Rationality calibration.** Fit β by locating confidence-shaping information peaks and testing the universal binary prediction $\beta_*|m| = 0.771702319\dots$
13. **Multiaction resonance geometry.** Generalize the boundary-layer profile from binary choice to arrangements of action-indifference hyperplanes.

A Derivation details for the multiclass envelope

A.1 Minimum KL at fixed top value

Let $q \in \Delta_K$ and $m = \max_i q_i = 1/K + v$. Holding m fixed, entropy is maximized by setting one coordinate to m and spreading the remaining mass equally over the other $K - 1$ coordinates. This gives

$$q^\uparrow(v) = \left(m, \frac{1-m}{K-1}, \dots, \frac{1-m}{K-1} \right), \quad (107)$$

with

$$D_{\text{KL}}(q^\dagger(v) \| u_K) = \Psi_K(v). \quad (108)$$

Since Ψ_K is increasing and convex, Jensen gives the mist upper envelope.

A.2 Maximum KL at fixed top value

Holding m fixed, entropy is minimized by making the posterior as concentrated as possible subject to every coordinate being at most m . Let $n = \lfloor 1/m \rfloor$ and $r = 1 - nm$. The extremal posterior has n coordinates equal to m , one coordinate equal to r if $r > 0$, and the rest zero. Its KL is

$$\Phi_K(m) = \log K + nm \log m + r \log r, \quad (109)$$

with the convention $0 \log 0 = 0$.

The upper concave envelope of Φ_K as a function of $v = m - 1/K$ is the polygon through uniform-support crystals

$$q^{(j)} = \left(\underbrace{1/j, \dots, 1/j}_{j \text{ entries}}, 0, \dots, 0 \right). \quad (110)$$

The corresponding points are $(V_j, I_j) = (1/j - 1/K, \log(K/j))$. The slope between adjacent crystals is

$$\frac{I_j - I_{j+1}}{V_j - V_{j+1}} = j(j+1) \log \left(1 + \frac{1}{j} \right), \quad (111)$$

which decreases as one moves from the uniform crystal $j = K$ to the one-point crystal $j = 1$. Hence the polygon is concave and supports the maximum-KL curve. This gives the lower envelope.

A.3 Achieving the crystals

To achieve the j -support crystal, draw Y uniformly from $\{1, \dots, K\}$. The signal S is a subset of size j containing Y , selected uniformly among all such subsets. Then $P(Y = i | S) = 1/j$ for $i \in S$ and zero otherwise. Thus

$$I(Y; S) = \log(K/j), \quad V_K(S) = 1/j - 1/K. \quad (112)$$

Randomizing between adjacent subset sizes achieves every point on the piecewise-linear lower envelope.

B Derivation details for the semantic eigenlens and compression identity

B.1 Eigenlens expansion

Let $q_s = p + \epsilon Z_s + O(\epsilon^2)$ with $\mathbb{E} Z_s = 0$ and $\mathbf{1}^\top Z_s = 0$. Taylor expansion gives

$$F(q_s) = F(p) + \epsilon \nabla F(p)^\top Z_s + \frac{\epsilon^2}{2} Z_s^\top \nabla^2 F(p) Z_s + O(\epsilon^3). \quad (113)$$

After averaging, the linear term vanishes. This gives

$$\mathcal{I}_F(Y; S) = \frac{\epsilon^2}{2} \mathbb{E} Z_s^\top \nabla^2 F(p) Z_s + O(\epsilon^3). \quad (114)$$

Similarly,

$$D_{\text{KL}}(p + \epsilon Z_s \| p) = \frac{\epsilon^2}{2} Z_s^\top D_p^{-1} Z_s + O(\epsilon^3), \quad (115)$$

so $I(Y; S) = \frac{\epsilon^2}{2} \mathbb{E} Z_S^\top D_p^{-1} Z_S + O(\epsilon^3)$. Setting $X_s = D_p^{-1/2} Z_s$ yields the stated Rayleigh quotient for $B_F(p)$.

B.2 Grouped-state spectrum

For the grouped-state utility, the action utility vectors are group indicators u_r . At the uniform prior, the Gibbs distribution over actions is uniform, so

$$\nabla^2 F_{u,\beta}(p) = \beta \left(\frac{1}{R} \sum_{r=1}^R u_r u_r^\top - \bar{u} \bar{u}^\top \right). \quad (116)$$

On perturbations that are uniform within each nuisance fiber, write $d_r = \sum_n Z_{r,n}$ with $\sum_r d_r = 0$. Then

$$Z^\top \nabla^2 F_{u,\beta}(p) Z = \frac{\beta}{R} \sum_r d_r^2, \quad (117)$$

whereas

$$Z^\top D_p^{-1} Z = R \sum_r d_r^2. \quad (118)$$

Thus the exchange rate is β/R^2 on the $R - 1$ relevant directions. If $d_r = 0$ for every group, the perturbation changes only nuisance conditionals and has zero ABI curvature. This gives the claimed spectrum.

B.3 Compression identity

The proof of the compression identity is included in the main text. Its algorithmic corollary follows because, for a fixed cell $C_t = \{s : g(s) = t\}$, the unique Bregman centroid minimizing $\sum_{s \in C_t} P(s) D_F(q_s, c)$ is the weighted mean $c = \sum_{s \in C_t} P(s | t) q_s = q_t$. Therefore an optimal deterministic M -message semantic compressor is a weighted Bregman M -means problem on posterior beliefs.

C Derivation details for action-policy geometry and rationality resonance

C.1 Policy-KL orientation

For log-partition functions, the Bregman divergence orientation is sometimes easy to reverse accidentally. With

$$A(\theta) = \log \sum_a \pi(a) e^{\theta \cdot u_a}, \quad (119)$$

one has

$$D_A(\theta_q, \theta_p) = A(\theta_q) - A(\theta_p) - \nabla A(\theta_p) \cdot (\theta_q - \theta_p) = D_{\text{KL}}(P_{\theta_p} \| P_{\theta_q}). \quad (120)$$

The first argument of the Bregman divergence becomes the second argument of the KL divergence. Therefore $D_F(q, p)$ compares the prior policy ω_p to the posterior policy ω_q in the order $D_{\text{KL}}(\omega_p \| \omega_q)$.

C.2 Binary soft value formula

For binary correct-prediction utility and belief $P(Y = 1) = 1/2 + x$,

$$F_\beta(x) = \frac{1}{\beta} \log \left(e^{\beta(1/2+x)} + e^{\beta(1/2-x)} \right) = \frac{1}{2} + \frac{\log 2}{\beta} + \frac{1}{\beta} \log \cosh(\beta x). \quad (121)$$

The constant terms cancel in all Jensen gaps, leaving $\varphi_\beta(x) = \beta^{-1} \log \cosh(\beta x)$. The derivatives are

$$\varphi'_\beta(x) = \tanh(\beta x), \quad \varphi''_\beta(x) = \beta \operatorname{sech}^2(\beta x), \quad (122)$$

which yields the resonance expansion.

C.3 Peak constants

The peak of the leading response $\beta \operatorname{sech}^2(\beta m)$ is obtained by setting $x = \beta|m|$ and solving

$$1 - 2x \tanh x = 0. \quad (123)$$

The unique positive root is

$$x_* = 0.771702319209104 \dots, \quad (124)$$

and the corresponding normalized height is

$$C_* = \frac{x_*}{2} \operatorname{sech}^2 x_* = 0.223871602347151 \dots \quad (125)$$

D New result: utility-score sufficiency and semantic dark information

The previous sections show that ABI is a Jensen gap over posterior beliefs and that its local curvature is controlled by action utility covariance. This section records a stronger structural fact. For soft action potentials, the full posterior $P_{Y|S}$ is not the primitive semantic object. The primitive object is the vector of posterior expected utilities for the available actions.

Let $K = |Y|$, $m = |\mathcal{A}|$, and let $U \in \mathbb{R}^{m \times K}$ be the utility matrix with entries $U_{ay} = u(a, y)$. Define the posterior utility-score map

$$M_U(p) = Up \in \mathbb{R}^m. \quad (126)$$

Then

$$F_{u,\beta}(p) = g_\beta(M_U(p)), \quad g_\beta(v) = \frac{1}{\beta} \log \sum_a \pi(a) e^{\beta v_a}. \quad (127)$$

Theorem 5 (Utility-score sufficiency). *For any signal S ,*

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) = \mathbb{E}_S g_\beta(UP_{Y|S}) - g_\beta(UP_Y).} \quad (128)$$

Consequently, any compression T satisfying

$$UP_{Y|T} = UP_{Y|S} \quad \text{a.s.} \quad (129)$$

preserves ABI exactly. In particular, $T_U(S) = UP_{Y|S}$ is a lossless semantic statistic for the action problem, even though it may discard arbitrary Shannon information about Y .

Proof. The definition of the soft action potential depends on p only through the expected utility vector Up . Substitution gives the identity. If $UP_{Y|T} = UP_{Y|S}$ almost surely, the two Jensen gaps are identical. $\square$

This theorem suggests an exact semantic compression principle: transmit the posterior action-score vector, not the full posterior, whenever the receiver’s utility model is known. The target dimension is at most $m - 1$ after removing additive constants common to all actions, independent of the number of world states.

D.1 Null-pair injection

The sufficiency theorem exposes an extreme separation between Shannon information and action value. A posterior displacement $z \in \mathbb{R}^K$ is utility-null if

$$\mathbf{1}^\top z = 0, \quad Uz = 0. \quad (130)$$

The first condition keeps the posterior normalized. The second condition keeps all posterior action scores fixed.

Theorem 6 (Null-pair injection). *Let an experiment contain an outcome of weight w with posterior q . Let z be utility-null and choose γ so that $q \pm \gamma z \in \Delta_Y$. Replace the outcome by two outcomes of weights $w/2, w/2$ and posteriors $q + \gamma z, q - \gamma z$. Then:*

$$\text{the prior is unchanged,} \quad \text{ABI is unchanged,} \quad (131)$$

while Shannon mutual information increases by

$$\Delta I = \frac{w}{2} D_{\text{KL}}(q + \gamma z \| p) + \frac{w}{2} D_{\text{KL}}(q - \gamma z \| p) - w D_{\text{KL}}(q \| p) \geq 0. \quad (132)$$

For small γ ,

$$\Delta I = \frac{w\gamma^2}{2} \sum_i \frac{z_i^2}{q_i} + \frac{w\gamma^4}{12} \sum_i \frac{z_i^4}{q_i^3} + O(\gamma^6). \quad (133)$$

Proof. The average posterior of the split pair is q , so the overall prior is unchanged. Since $U(q + \gamma z) = Uq = U(q - \gamma z)$, the split outcomes have the same action-score vector as the original outcome, and the ABI contribution is unchanged by utility-score sufficiency. The mutual-information increase is the Jensen gap of the convex map $r \mapsto D_{\text{KL}}(r \| p)$. The expansion follows from the Taylor series of $(q_i \pm \gamma z_i) \log((q_i \pm \gamma z_i)/p_i)$ and cancellation of odd powers. $\square$

This result may be called a semantic-dark-information law: in the presence of utility-null directions, raw mutual information can be injected into a signal without changing its action value.

D.2 Action-rank bottleneck

The same structure gives a local rank bound.

Theorem 7 (Action-rank bottleneck). *For the soft action potential,*

$$\nabla^2 F_{u,\beta}(p) = \beta U^\top \left(\text{diag } \omega(p) - \omega(p) \omega(p)^\top \right) U. \quad (134)$$

Hence

$$\boxed{\text{rank } \nabla^2 F_{u,\beta}(p) \leq \min\{K-1, m-1, \text{rank } U\}.} \quad (135)$$

When the state space is much larger than the action set, almost all local posterior directions are invisible to ABI to second order.

Proof. The Hessian formula is the decision-curvature law written in matrix form. The action covariance matrix $\text{diag } \omega - \omega\omega^\top$ has rank at most $m-1$, and multiplication by $U^\top$ and U cannot increase rank. The simplex tangent itself has dimension $K-1$. $\square$

D.3 Isotropic semantic visible fraction

Let $p_i = 1/K$ and consider a weak binary experiment with posteriors $p \pm \varepsilon Z$, where Z is drawn isotropically from the tangent sphere $\mathbf{1}^\top Z = 0$. Let $\lambda_1, \dots, \lambda_{K-1}$ be the generalized semantic eigenvalues

$$\nabla^2 F(p)v_i = \lambda_i \text{diag}(1/p)v_i. \quad (136)$$

Then

$$\boxed{\mathbb{E}_Z \left[\frac{\mathcal{I}_F(Y; S_Z)}{I(Y; S_Z)} \right] = \frac{1}{K-1} \sum_{i=1}^{K-1} \lambda_i + O(\varepsilon).} \quad (137)$$

For a group task $Y = (R, N)$ with R relevant classes and G nuisance states per class, uniform prior, and correct-class utility, the generalized eigenvalues are

$$\lambda = \frac{\beta}{R^2} \quad \text{with multiplicity } R-1, \quad \lambda = 0 \quad \text{with multiplicity } RG-R. \quad (138)$$

Therefore

$$\boxed{\mathbb{E} \left[\frac{\mathcal{I}_{u,\beta}}{I} \right] = \frac{\beta(R-1)}{R^2(RG-1)} + O(\varepsilon).} \quad (139)$$

Thus the useful fraction of random low-rate Shannon information decays like G^{-1} as nuisance dimension grows.

D.4 Numerical tests

A null-pair injection experiment used $K = 18$ world states, $m = 4$ actions, and $\beta = 3$. The utility-null space had dimension 13. Increasing the normalized null-injection amplitude from 0 to 1 changed Shannon mutual information from 0.0111115 to 0.0738159, while ABI remained 0.003960203 with maximum numerical drift 8.8×10^{-17} .

The action-rank experiment used $K = 80$ states and random utility matrices. The measured local semantic ranks were exactly $m-1$ for $m = 2, 3, 4, 8, 16, 32$ actions.

The group-task visible-fraction experiment used $R = 4$, $\beta = 2$, and 2500 random weak binary channels for each nuisance size G . The measured mean ABI/MI ratios tracked the exact isotropic prediction

$$\frac{2(4-1)}{4^2(4G-1)}. \quad (140)$$

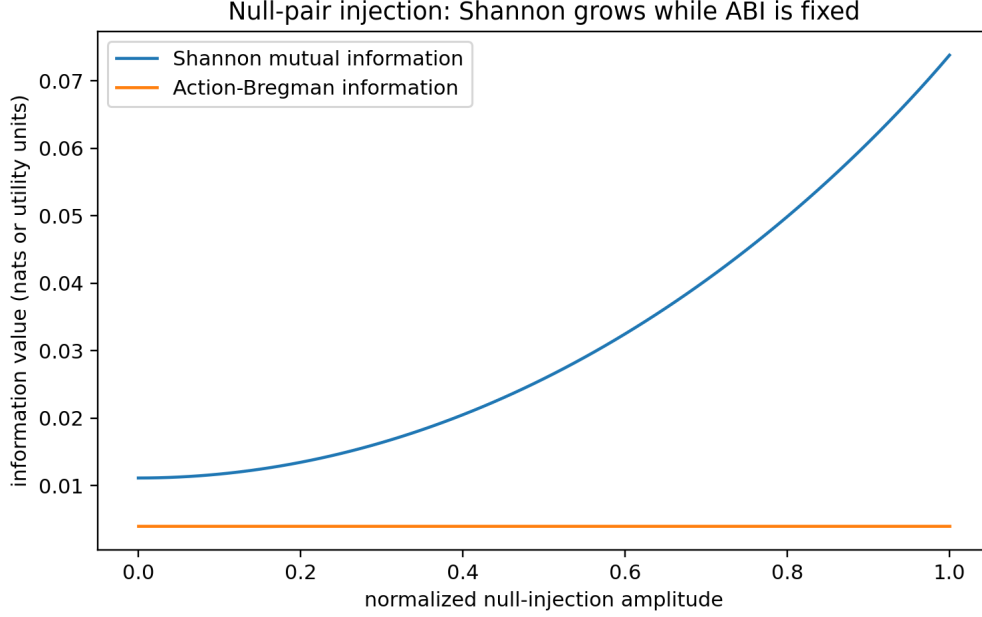


Figure 17: Null-pair injection increases Shannon mutual information while preserving ABI to roundoff. The injected information lies in directions that do not alter posterior action-score vectors.

D.5 Applications and targets opened by v0.6

- **Lossless semantic compression.** The statistic $UP_{Y|S}$ is an exact ABI-preserving representation. It can be much lower-dimensional than a posterior over Y .
- **Privacy and leakage audits.** Null-pair injection separates raw leakage from task-relevant leakage. A system can leak many Shannon nats while leaking no value for a specified action model, or conversely leak little Shannon information in a highly action-aligned direction.
- **Representation learning.** The action-rank bottleneck suggests a regularizer that discards posterior directions lying in $\ker U$ or in low semantic eigenvalue directions.
- **Semantic communication.** Bandwidth should be allocated to action-score coordinates rather than nuisance coordinates. In high-dimensional scenes, isotropic compression wastes nearly all low-rate capacity unless the representation is semantically aligned.
- **Null capacity.** Define the semantic-dark capacity

$$C_{\text{null}}(p, U) = \sup\{I(Y; S) : \mathcal{I}_{u, \beta}(Y; S) = 0\}. \quad (141)$$

For partition utilities with independent nuisance variables, this equals the nuisance entropy. For general utility matrices, it becomes a new convex-geometric optimization over posterior decompositions inside fibers of $p \mapsto Up$.

E New result: semantic-dark capacity and fiber polytopes

The v0.6 null-pair construction showed that Shannon information can be injected along directions invisible to the action-score map. This section sharpens that observation from a local construction

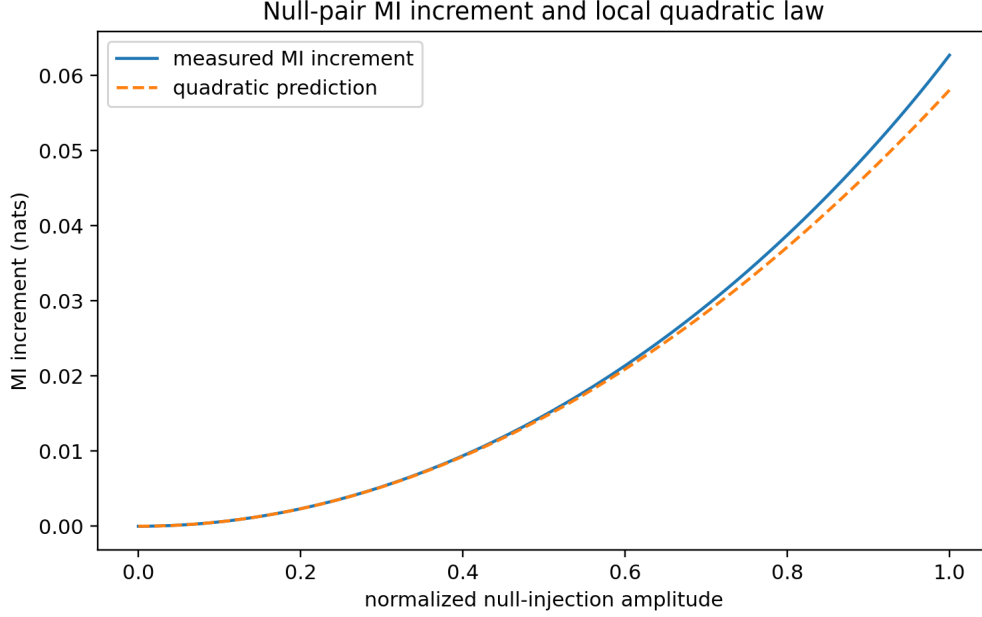


Figure 18: Measured null-pair MI increment compared with the local quadratic prediction. Deviations at large amplitude are expected from higher even powers.

into a capacity theorem. The central object is not the full utility matrix U , but its quotient after removing action-independent score shifts.

E.1 Action-gauge invariance

Let $U \in \mathbb{R}^{m \times K}$ be the utility matrix. For any vector $b \in \mathbb{R}^K$, define a gauged utility matrix

$$U' = U + \mathbf{1}_m b^\top, \quad (142)$$

where $\mathbf{1}_m$ is the all-ones vector over actions. Then

$$F_{U',\beta}(p) = b^\top p + F_{U,\beta}(p). \quad (143)$$

Because posterior beliefs average to the prior, the linear term cancels in the Jensen gap. Hence

$$\boxed{\mathcal{I}_{U',\beta}(Y; S) = \mathcal{I}_{U,\beta}(Y; S).} \quad (144)$$

Action-independent state reward is therefore a semantic gauge: it may carry Shannon information, but it does not alter the relative value of actions.

Let $Q \in \mathbb{R}^{(m-1) \times m}$ be any full-rank quotient map with

$$\ker Q = \text{span}\{\mathbf{1}_m\}, \quad (145)$$

and define the quotient utility matrix

$$B = QU. \quad (146)$$

Only Bp matters for strict action-relevant curvature. The rank

$$r = \text{rank } B \quad (147)$$

will be called the *action-score rank*.

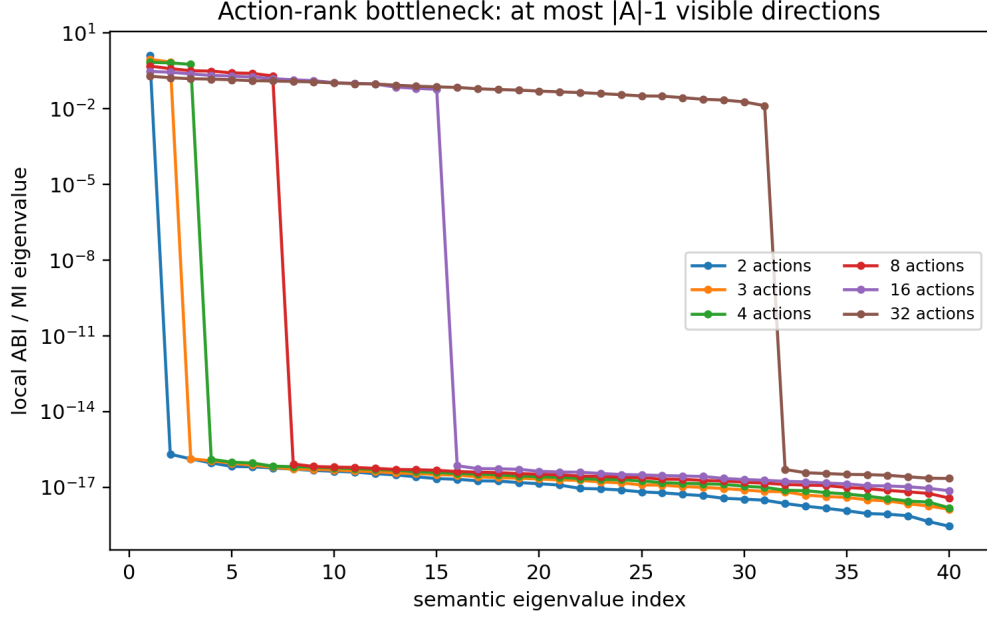


Figure 19: Semantic eigenvalue spectra for random utility matrices. The sharp drop after $m - 1$ eigenvalues confirms the action-rank bottleneck.

Theorem 8 (Zero-ABI fiber criterion). *Assume the reference action prior has full support. Let $p = P_Y$ and $q_S = P_{Y|S}$. Then*

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) = 0 \iff Bq_S = Bp \text{ a.s.}} \quad (148)$$

for every $\beta > 0$.

Proof. Let $v_S = Uq_S$ and $\bar{v} = Up$. The log-sum-exp potential $g_\beta(v)$ is convex, and its Hessian is the covariance matrix of action indicators under the Gibbs action distribution. With full-support action prior, this Hessian has nullspace exactly $\text{span}\{\mathbf{1}_m\}$ and is positive definite on the quotient space $\mathbb{R}^m / \text{span}\{\mathbf{1}_m\}$. Jensen equality therefore holds exactly when $v_S - \bar{v}$ lies in $\text{span}\{\mathbf{1}_m\}$ almost surely. Applying Q gives $Bq_S = Bp$. The converse follows because $g_\beta(v + c\mathbf{1}_m) = g_\beta(v) + c$, and the affine shifts c average to zero since $\mathbb{E}v_S = \bar{v}$. $\square$

Thus zero action value does not mean zero information. It means that all posterior beliefs remain inside a fixed affine fiber of the quotient score map.

E.2 Dark capacity formula

Define the zero-ABI fiber through the prior p by

$$\mathcal{F}_B(p) = \{q \in \Delta_K : Bq = Bp\}. \quad (149)$$

The *semantic-dark capacity* is the largest Shannon mutual information attainable by an experiment whose ABI is exactly zero:

$$\boxed{C_0(p, U) = \sup\{I(Y; S) : \mathcal{I}_{u,\beta}(Y; S) = 0\}.} \quad (150)$$

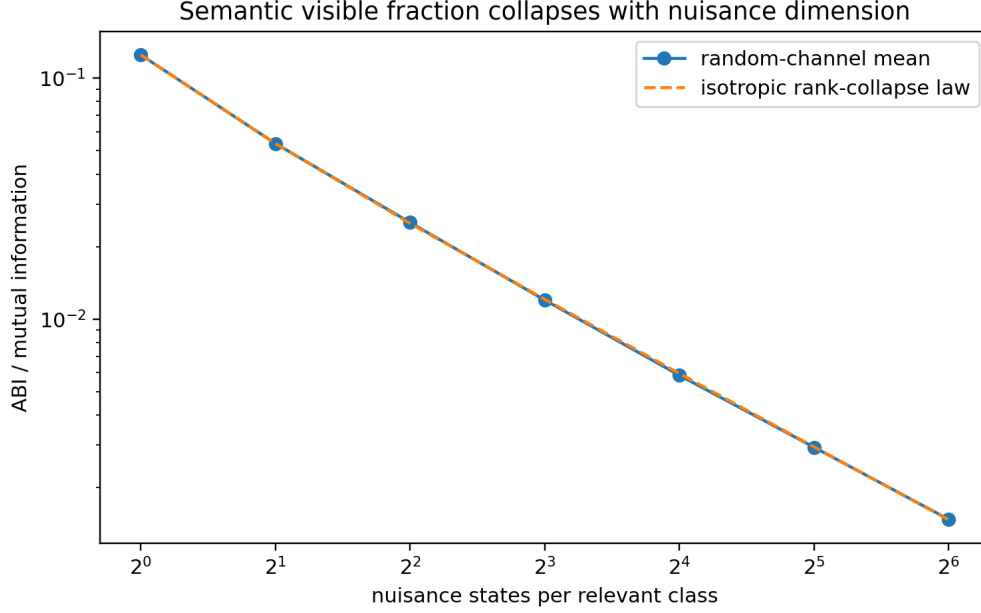


Figure 20: In nuisance-heavy state spaces, random Shannon information is almost entirely semantic-null. The useful fraction follows the predicted $1/G$ collapse.

By the zero-ABI fiber criterion, C_0 is independent of β and depends on U only through the quotient matrix B .

Let

$$\underline{H}_B(p) = \inf \left\{ \sum_i w_i H(q_i) : q_i \in \mathcal{F}_B(p), w_i \geq 0, \sum_i w_i = 1, \sum_i w_i q_i = p \right\}. \quad (151)$$

This is the lower convex envelope of entropy restricted to the fiber, evaluated at p .

Theorem 9 (Fiber-entropy formula).

$$\boxed{C_0(p, U) = H(p) - \underline{H}_B(p)}. \quad (152)$$

Moreover, the infimum may be achieved using finitely many posterior points in $\mathcal{F}_B(p)$.

Proof. Any finite experiment is equivalent to a probability distribution over posterior beliefs q_i whose average is the prior p . Its mutual information is

$$I(Y; S) = H(p) - \sum_i w_i H(q_i). \quad (153)$$

The zero-ABI condition is precisely $q_i \in \mathcal{F}_B(p)$ for all outcomes. Maximizing mutual information is therefore the same as minimizing average posterior entropy inside this fiber. Finite attainment follows from compactness and Caratheodory reduction. $\square$

E.3 Dark simplex law

The fiber $\mathcal{F}_B(p)$ is a polytope. It is defined by nonnegativity, normalization, and r independent quotient-score constraints. Therefore every vertex of the fiber has support size at most $r + 1$.

Theorem 10 (Support bound and dark simplex law). *Let $r = \text{rank } B$. Then*

$$\boxed{C_0(p, U) \geq [H(p) - \log(r + 1)]_+,} \quad (154)$$

where $[x]_+ = \max\{x, 0\}$. *In particular, under the uniform prior on K states,*

$$\boxed{C_0(p, U) \geq \log \frac{K}{r + 1} \quad \text{whenever } K > r + 1.} \quad (155)$$

Proof. Decompose p as a convex combination of vertices of $\mathcal{F}_B(p)$. Each vertex has support at most $r + 1$, hence entropy at most $\log(r + 1)$. Thus $\underline{H}_B(p) \leq \log(r + 1)$ and the bound follows. $\square$

This is an unexpectedly strong blind-spot law. If the state space is large but the action-score rank is small, then a large number of Shannon nats can be made completely invisible to ABI. The limit is not set by the state alphabet; it is set by the dimension of the action quotient.

E.4 Exact partition-task capacity

Let $Y = (R, N)$, where R is the decision-relevant class and N is a nuisance state. Suppose actions are class predictions and the utility depends only on R . Then the quotient-score fiber fixes the posterior marginal over R and leaves the conditional nuisance distribution free.

Theorem 11 (Partition dark capacity). *For any prior $p(r, n)$ and correct-class utility depending only on R ,*

$$\boxed{C_0(p, U) = H(N \mid R).} \quad (156)$$

For a uniform prior over R classes and G nuisance states per class,

$$\boxed{C_0 = \log G.} \quad (157)$$

Proof. The fiber condition fixes $q_R = p_R$. Therefore every posterior in the fiber has entropy

$$H(q) = H(p_R) + H_q(N \mid R) \geq H(p_R). \quad (158)$$

This gives $\underline{H}_B(p) \geq H(p_R)$. Conversely, for each function f selecting one nuisance state $f(r)$ for every class r , define a posterior concentrated at $(r, f(r))$ with class marginal p_R . By mixing such posteriors with a coupling whose marginals reproduce $p(n \mid r)$, one obtains the prior p . Each selected posterior has entropy exactly $H(p_R)$, so $\underline{H}_B(p) = H(p_R)$. Hence $C_0 = H(p) - H(p_R) = H(N \mid R)$. $\square$

Thus a signal can reveal the full nuisance coordinate conditionally on the relevant class while preserving zero action value. This is not a failure of Shannon information; it is a separation between truthful state information and decision-visible information.

E.5 Numerical experiments

The experiments for this section used exact vertex enumeration of the fiber polytope for moderate state sizes, followed by a linear program over vertices to minimize average posterior entropy. For a fiber with quotient rank r , candidate vertices were enumerated over supports of size at most $r + 1$.

For partition tasks with $R \leq 4$ and $G \leq 4$, exact enumeration verified $C_0 = \log G$ with maximum absolute residual 4.44×10^{-16} . The plotted curves show the formula up to $G = 12$.

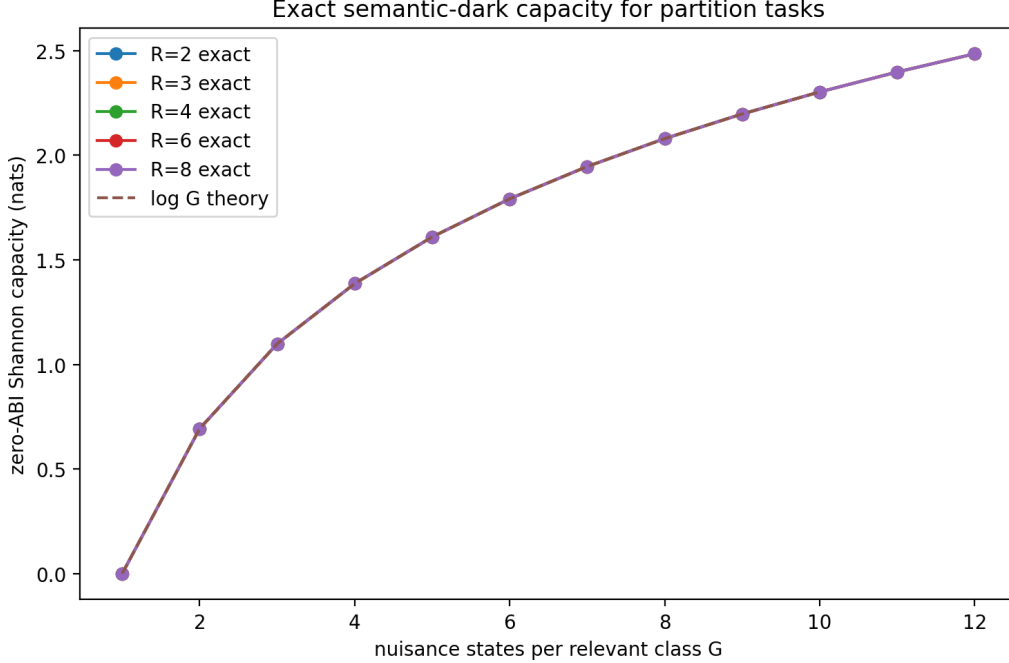


Figure 21: Semantic-dark capacity for partition tasks. For uniform class-nuisance worlds, the zero-ABI Shannon capacity is exactly $\log G$, independent of the number of relevant classes R .

Random utility experiments used uniform priors and Gaussian utility matrices. A random state-dependent baseline was added to each action as a gauge test. Across all tested random instances, the maximum absolute capacity change under this gauge transformation was 3.44×10^{-14} .

A representative case with $K = 16$ states and $m = 4$ actions had quotient rank $r = 3$, affine rank $r + 1 = 4$, and 304 fiber vertices. The exact linear program found

$$C_0 = 1.7366989723 \text{ nats}, \quad H(p) - \log 4 = 1.3862943611 \text{ nats}. \quad (159)$$

The optimal decomposition used 13 vertices and had average posterior entropy 1.0358897500 nats.

The channel produced by this optimal vertex decomposition was then evaluated directly for several inverse rationalities β . Its Shannon mutual information equaled the computed dark capacity, while ABI stayed at roundoff error for all tested β :

$$\max_{\beta \in \{0.5, 1, 2, 5, 10, 50\}} |\mathcal{I}_\beta| = 1.67 \times 10^{-16}. \quad (160)$$

E.6 Implications and applications

- **Semantic privacy audits.** Leakage should be decomposed into action-visible and action-dark parts. A large Shannon leakage may be harmless for one specified decision model but highly consequential for another.
- **Task-aware telemetry.** Sensors may be designed to record high-volume nuisance detail while guaranteeing zero effect on a downstream policy, provided the quotient-score fiber constraint is enforced.

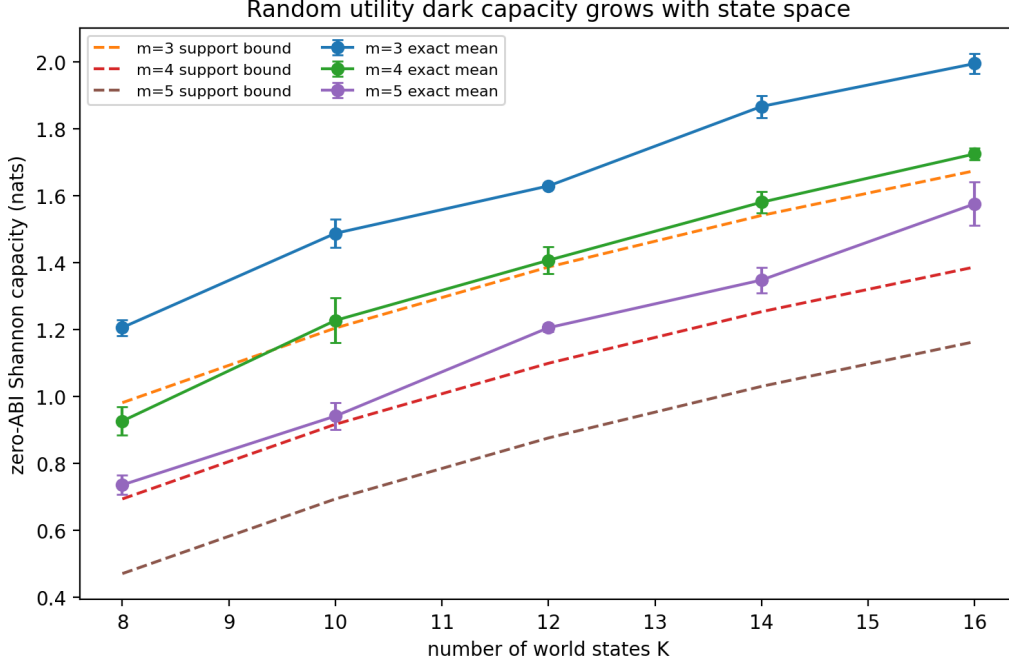


Figure 22: Exact semantic-dark capacity for random utility matrices. The dashed lines show the support bound $H(p) - \log(r + 1)$. The exact capacity generally exceeds the bound because many fiber vertices have entropy strictly below $\log(r + 1)$.

- **Representation learning.** The quotient matrix $B = QU$ identifies directions that must be preserved for action value. Orthogonal fiber directions can be compressed, masked, or separately routed.
- **Semantic communication.** When bandwidth is scarce, dark capacity quantifies how much raw detail can be transmitted without changing the receiver’s action potential. Conversely, it exposes when high-rate transmission is semantically inert.
- **Decision-system robustness.** The gauge law warns that action-independent state information can accumulate inside data streams without appearing in ABI metrics. Multi-utility audits are needed when future tasks may differ from the present utility model.

E.7 Targets for further study

1. **ϵ -dark capacity.** Extend C_0 to

$$C_\epsilon(p, U) = \sup\{I(Y; S) : \mathcal{I}_{u, \beta}(Y; S) \leq \epsilon\}, \quad (161)$$

and determine whether its expansion above C_0 is governed by quotient-score curvature at fiber vertices.

2. **Large-scale vertex algorithms.** Exact support enumeration is exponential in r . Develop column-generation, hit-and-run, and entropy-cutting algorithms for large state spaces.
3. **Multi-utility dark capacity.** For several possible downstream utilities $U_1, \dots, U_L$, the dark fiber is the intersection of quotient-score fibers. Its capacity should shrink according to the rank of the stacked quotient matrix.

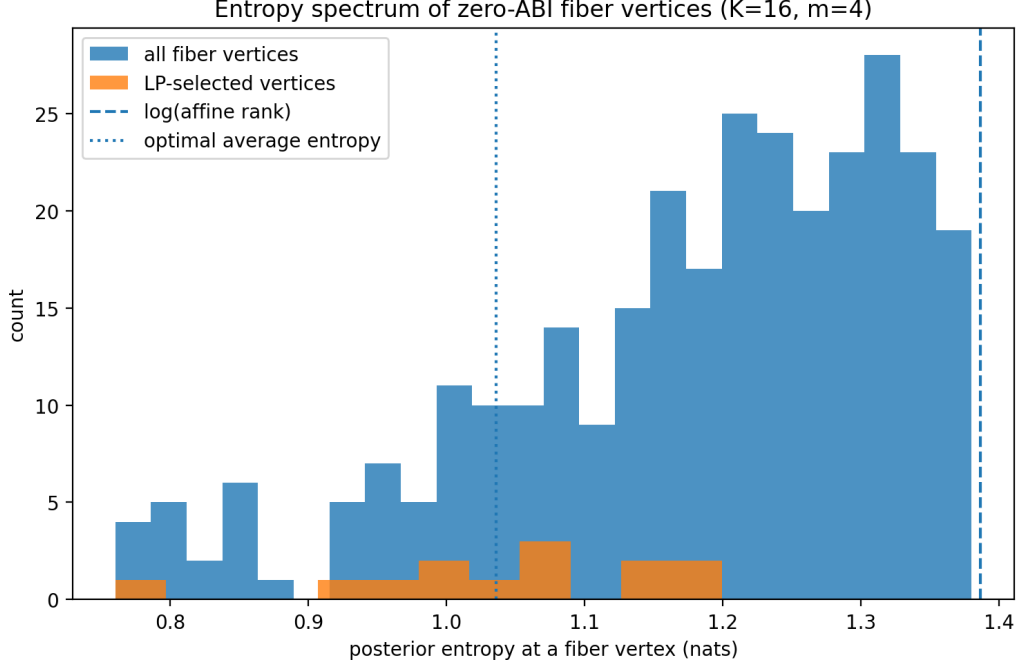


Figure 23: Entropy distribution of zero-ABI fiber vertices for the representative random case. The capacity-achieving decomposition selects low-entropy vertices, pushing the dark capacity above the universal support bound.

4. **Directed dark information.** Define a causal version in which $B_t P_{Y_t|Y^{t-1}, S^t} = B_t P_{Y_t|Y^{t-1}}$ at every time. This would separate causal Shannon flow from directed action value.
5. **Blackwell geometry of fibers.** Characterize when two experiments are equivalent for a given utility family even though they are not Blackwell equivalent for all utilities.

F Semantic water-filling under a Shannon budget

A natural continuation of the preceding dark-capacity theory is the bright allocation problem: when a finite amount of Shannon information can be acquired, how should it be distributed across decision coordinates? This section treats a tractable case that already shows a sharp phenomenon. Finite rationality causes low-rate information to freeze onto the strongest decision coordinate, while hard rationality spreads arbitrarily small information across all positive-value coordinates.

F.1 Weighted binary decision coordinates

Consider d independent balanced binary coordinates. Coordinate i has value weight $a_i > 0$. If its posterior margin is

$$t_i = \left| P(Y_i = 1 | S) - \frac{1}{2} \right|, \quad (162)$$

then the Shannon information carried by a symmetric mist signal is

$$\phi(t) = \left(\frac{1}{2} + t \right) \log(1 + 2t) + \left(\frac{1}{2} - t \right) \log(1 - 2t), \quad 0 \leq t \leq \frac{1}{2}. \quad (163)$$

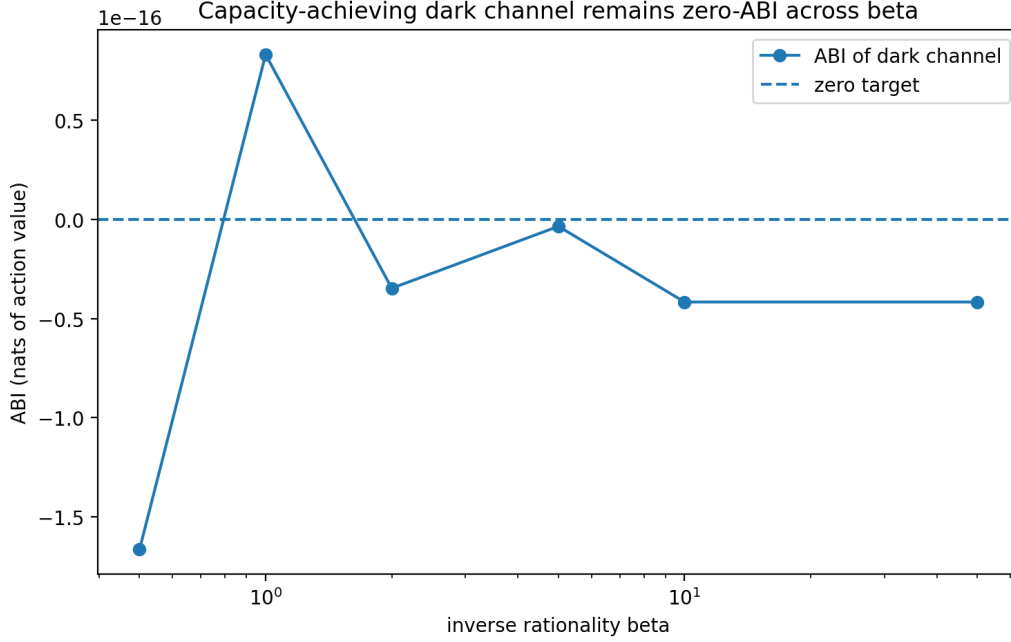


Figure 24: Direct certificate for a capacity-achieving dark channel in the representative random case. The constructed channel carries 1.7367 Shannon nats but has zero ABI across the tested rationality range.

For finite inverse rationality β , the coordinate value as a function of the coordinate information $I_i = \phi(t_i)$ is

$$U_i(I_i; \beta) = \frac{1}{\beta} \log \cosh(\beta a_i \phi^{-1}(I_i)). \quad (164)$$

The hard-rational limit is

$$U_i(I_i; \infty) = a_i \phi^{-1}(I_i). \quad (165)$$

The allocation problem is

$$V_\beta^*(B) = \max_{0 \leq I_i \leq \log 2, \sum_i I_i \leq B} \sum_{i=1}^d U_i(I_i; \beta). \quad (166)$$

F.2 Conductance and water-filling

Differentiating through $I_i = \phi(t_i)$ gives the semantic conductance of coordinate i :

$$\frac{dU_i}{dI_i} = s_i(t_i) = \frac{a_i \tanh(\beta a_i t_i)}{\log((1 + 2t_i)/(1 - 2t_i))}. \quad (167)$$

The finite-rational initial conductance is

$$s_i(0) = \frac{\beta a_i^2}{4}. \quad (168)$$

For $\beta = \infty$,

$$s_i^\infty(t) = \frac{a_i}{\log((1+2t)/(1-2t))}, \quad (169)$$

which diverges as $t \downarrow 0$.

Theorem 12 (Semantic water-filling). *For finite β , the optimal allocation is characterized by a shadow price $\nu \geq 0$ satisfying*

$$I_i^*(B) = \phi(t_i(\nu)), \quad s_i(t_i(\nu)) = \nu, \quad (170)$$

with endpoint conventions $t_i(\nu) = 0$ if $\nu \geq \beta a_i^2/4$ and $t_i(0) = 1/2$. The hard-rational case is obtained by replacing s_i with s_i^∞ .

Thus information fills coordinates until all active coordinates have equal marginal action value per nat. This is the decision-theoretic analogue of physical water-filling.

F.3 Freeze-out and noncommuting limits

Assume $a_1 > a_2 > \dots > a_d > 0$. At fixed finite β , the second coordinate activates only once the first coordinate conductance has fallen to $\beta a_2^2/4$, and similarly for later coordinates. Thus, for all sufficiently small $B > 0$,

$$I_1^*(B; \beta) = B, \quad I_i^*(B; \beta) = 0 \quad (i > 1). \quad (171)$$

In the hard-rational limit, all positive-value coordinates activate immediately, and

$$I_i^*(B; \infty) = \frac{a_i^2}{\sum_k a_k^2} B + O(B^2). \quad (172)$$

Therefore the low-rate and high-rationality limits do not commute:

$$\lim_{\beta \rightarrow \infty} \lim_{B \downarrow 0} \frac{I_i^*(B; \beta)}{B} = \mathbf{1}_{\{i=1\}}, \quad \lim_{B \downarrow 0} \lim_{\beta \rightarrow \infty} \frac{I_i^*(B; \beta)}{B} = \frac{a_i^2}{\sum_k a_k^2}. \quad (173)$$

The corresponding low-rate values are

$$V_\beta^*(B) = \frac{\beta a_1^2}{4} B + O(B^2) \quad (\beta < \infty), \quad (174)$$

whereas

$$V_\infty^*(B) = \sqrt{\frac{B}{2}} \left(\sum_i a_i^2 \right)^{1/2} + O(B^{3/2}). \quad (175)$$

F.4 Activation boundary layer

Let coordinate j activate after coordinates $1, \dots, j-1$. For large β , define $x_{ij} > 0$ by

$$\frac{\tanh x_{ij}}{x_{ij}} = \left(\frac{a_j}{a_i} \right)^2, \quad i < j. \quad (176)$$

Then

$$B_j(\beta) = \frac{C_j}{\beta^2} + O(\beta^{-4}), \quad C_j = 2 \sum_{i < j} \frac{x_{ij}^2}{a_i^2}. \quad (177)$$

This β^{-2} law is the boundary layer through which the hard-rational spreading solution emerges from finite-rational freeze-out.

F.5 Experiments

The numerical experiment used weights

$$(a_1, a_2, a_3, a_4, a_5) = (1.00, 0.78, 0.55, 0.35, 0.20). \quad (178)$$

At $\beta = 20$, the activation budgets were 0.010799, 0.080861, 0.503633, and 1.982593 nats for coordinates 2 through 5. The boundary-layer constants C_j were 4.395364, 33.955331, 253.437684, and 2540.378594.

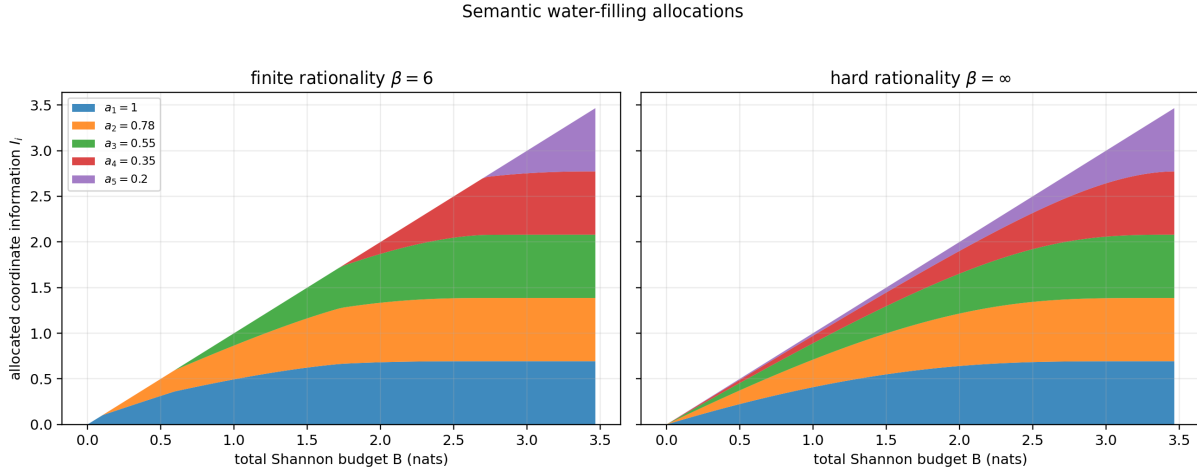


Figure 25: Optimal allocation of a Shannon budget across weighted binary decision coordinates. Finite rationality initially concentrates rate on the strongest coordinate; hard rationality spreads immediately.

Random allocation stress tests found that sampled feasible allocations remained below the KKT value. Across the tested budgets, the smallest observed random gap was 0.001300 and the largest was 0.021772.

Applications. Semantic water-filling gives a design rule for task-aware sensing, semantic communication, model compression, safety triage, and adaptive measurement: allocate nats by marginal action conductance rather than raw entropy.

G Semantic refraction: channel-independent efficiency caps

The water-filling law allocates nats once coordinate conductance curves are known. This section asks a more primitive question: how efficiently can Shannon information be converted into ABI for

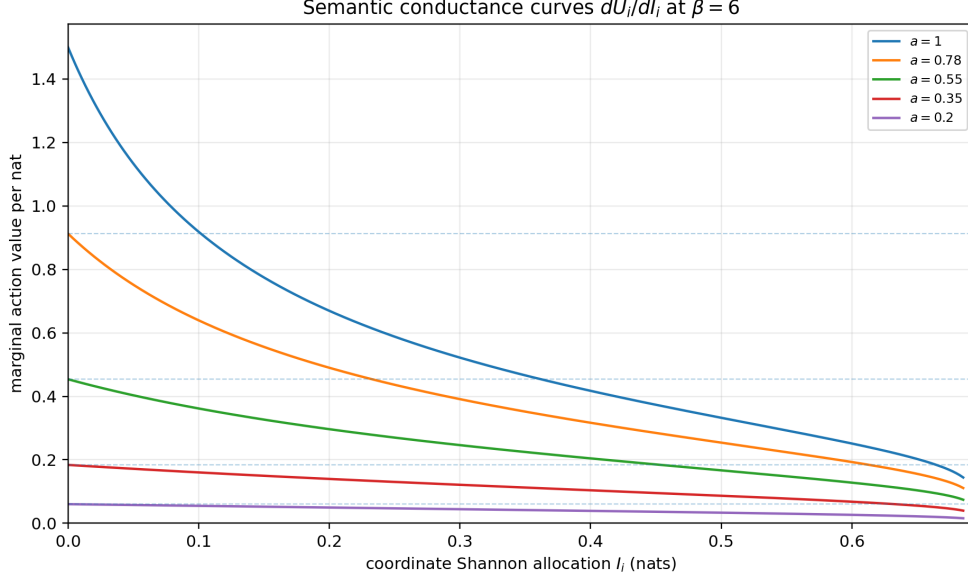


Figure 26: Semantic conductance curves at $\beta = 6$. Activation occurs when the common shadow price drops below a coordinate's initial conductance $\beta a_i^2/4$.

a fixed utility? The answer is a relative-smoothness law. At finite rationality, every utility imposes a channel-independent cap

$$\mathcal{I}_{u,\beta}(Y; S) \leq \Gamma_{u,\beta} I(Y; S), \quad (179)$$

where $\Gamma_{u,\beta}$ is the maximum local curvature of the soft action potential measured in the Fisher/Shannon metric. This is a semantic analogue of an index of refraction: Shannon nats entering different belief regions convert to action value at different rates.

G.1 Relative entropy smoothness

Let

$$\Phi(p) = \sum_y p_y \log p_y \quad (180)$$

be negative entropy. Then

$$I(Y; S) = \mathbb{E}_S D_\Phi(P_{Y|S}, P_Y). \quad (181)$$

For a convex potential F , ABI is

$$\mathcal{I}_F(Y; S) = \mathbb{E}_S D_F(P_{Y|S}, P_Y). \quad (182)$$

Theorem 13 (Relative-smoothness ABI cap). *Suppose F is twice differentiable on the simplex interior and satisfies the Hessian domination*

$$v^T \nabla^2 F(p) v \leq \Gamma \sum_y \frac{v_y^2}{p_y} \quad (183)$$

for every interior belief p and every tangent vector v with $\sum_y v_y = 0$. Then every finite signal satisfies

$$\boxed{\mathcal{I}_F(Y; S) \leq \Gamma I(Y; S)}. \quad (184)$$

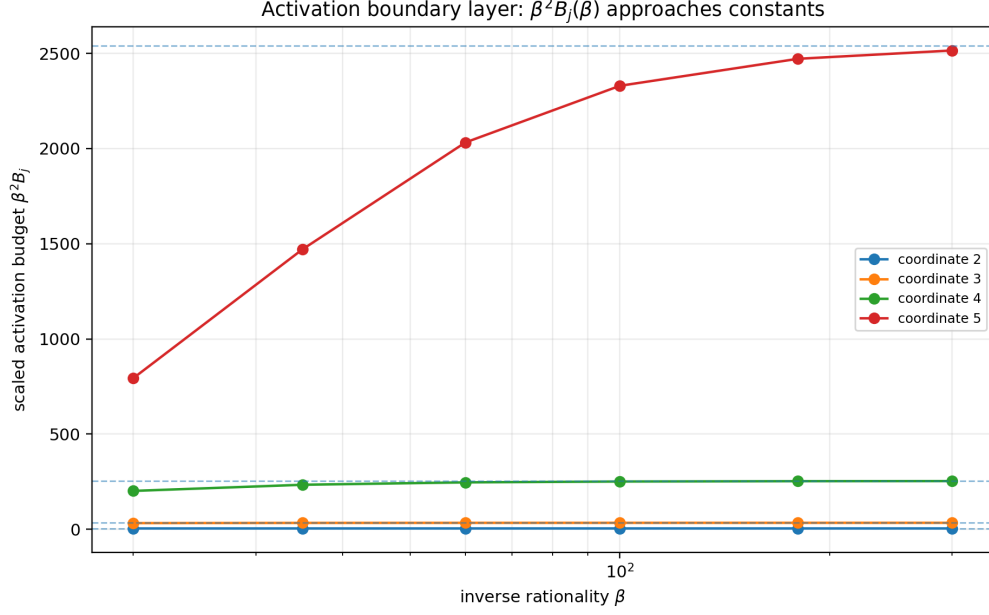


Figure 27: Activation boundary-layer test. The dashed lines are the predicted constants C_j .

Proof sketch. The Hessian condition says that F is Γ -smooth relative to Φ . Integrating the second derivative along the line segment from p to q gives $D_F(q, p) \leq \Gamma D_\Phi(q, p)$. Averaging with $p = P_Y$ and $q = P_{Y|S}$ gives the result. $\square$

G.2 The semantic refraction coefficient

For the soft action potential

$$F_{u,\beta}(p) = \frac{1}{\beta} \log \sum_a \pi(a) \exp\{\beta(Up)_a\}, \quad (185)$$

where $U_{ay} = u(a, y)$, the Hessian is

$$\nabla^2 F_{u,\beta}(p) = \beta U^T (\text{diag } \omega_p - \omega_p \omega_p^T) U, \quad (186)$$

where

$$\omega_p(a) = \frac{\pi(a) e^{\beta(Up)_a}}{\sum_b \pi(b) e^{\beta(Up)_b}}. \quad (187)$$

Define the local semantic refraction coefficient

$$\Gamma_{u,\beta}(p) = \lambda_{\max} \left(P_p D_p^{1/2} \nabla^2 F_{u,\beta}(p) D_p^{1/2} P_p \right), \quad (188)$$

where $D_p = \text{diag}(p)$ and $P_p = I - \sqrt{p} \sqrt{p}^T$ projects onto the tangent subspace in Fisher coordinates. The global coefficient is

$$\Gamma_{u,\beta} = \sup_{p \in \Delta^\circ} \Gamma_{u,\beta}(p). \quad (189)$$



Figure 28: Value under optimal water-filling compared with uniform allocation and a naive fill-by-weight rule.

The relative-smoothness theorem gives

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) \leq \Gamma_{u,\beta} I(Y; S).} \quad (190)$$

This coefficient is the exact local exchange rate between Shannon nats and action value. More precisely, if v is the maximizing tangent direction at p and a binary signal has posteriors $q_{\pm} = p \pm \epsilon v$, then

$$\boxed{\frac{\mathcal{I}_{u,\beta}(Y; S_{\epsilon})}{I(Y; S_{\epsilon})} \longrightarrow \Gamma_{u,\beta}(p) \quad (\epsilon \downarrow 0).} \quad (191)$$

Thus $\Gamma_{u,\beta}(p)$ is not merely a bound; it is an attainable infinitesimal efficiency.

G.3 Universal action-gap cap

The exact global coefficient can require optimization over the belief simplex. A simple closed-form certificate is available. Define the statewise action-gap diameter

$$\Delta_{\infty} = \max_{a,b,y} |u(a, y) - u(b, y)|. \quad (192)$$

Then

$$\boxed{\Gamma_{u,\beta} \leq \frac{\beta \Delta_{\infty}^2}{4}.} \quad (193)$$

Consequently

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) \leq \frac{\beta \Delta_{\infty}^2}{4} I(Y; S).} \quad (194)$$

Proof sketch. For a tangent vector v ,

$$v^T \nabla^2 F v = \beta \text{Var}_{a \sim \omega_p}(u_a \cdot v). \quad (195)$$

By the range variance bound,

$$\text{Var}_a(u_a \cdot v) \leq \frac{1}{4} \max_{a,b} ((u_a - u_b) \cdot v)^2. \quad (196)$$

Cauchy's inequality gives

$$((u_a - u_b) \cdot v)^2 \leq \left(\sum_y p_y (u(a, y) - u(b, y))^2 \right) \left(\sum_y \frac{v_y^2}{p_y} \right) \leq \Delta_\infty^2 \sum_y \frac{v_y^2}{p_y}. \quad (197)$$

Combining these inequalities gives the claim. $\square$

For binary correct-prediction utility, $\Delta_\infty = 1$, so this recovers the soft semantic efficiency cap $\mathcal{I}_\beta \leq (\beta/4)I$ obtained earlier. The new result shows that the cap is not special to binary prediction: finite rationality generally prevents arbitrarily large useful value per nat unless utilities have unbounded action gaps.

G.4 Numerical experiments

Three experiments were run.

First, 30 random utility matrices with 7 states and 4 actions were generated across $\beta \in \{1.5, 3, 6, 10\}$. For each matrix, $\Gamma_{u,\beta}$ was estimated by sampling beliefs from several Dirichlet distributions, and 180 random finite channels were tested. Across 5400 random channels, the largest residual against the certified action-gap cap was

$$\max(\mathcal{I} - (\beta \Delta_\infty^2/4)I) = -0.003758, \quad (198)$$

so no violation was observed. The maximum random-channel ratio divided by the sampled local cap was 0.654588, consistent with random channels rarely aligning with the most efficient local eigenchannel.

Second, the universal action-gap cap was compared with the sampled task-specific curvature cap. The median ratio $(\beta \Delta_\infty^2/4)/\hat{\Gamma}$ was 3.438156, and the maximum was 15.870983. Thus the closed-form cap is useful as a certificate but often loose; task geometry can lower the real semantic efficiency by more than an order of magnitude.

Third, a representative utility was used to construct the local maximizing eigenchannel. The predicted local coefficient was

$$\Gamma_{u,\beta}(p) = 0.6190121220908867. \quad (199)$$

For posterior perturbation sizes from 10^{-4} to 10^{-2} of the positivity radius, the measured ratio $\mathcal{I}/I$ agreed with this value to the shown numerical precision, then drifted downward for finite-amplitude channels.

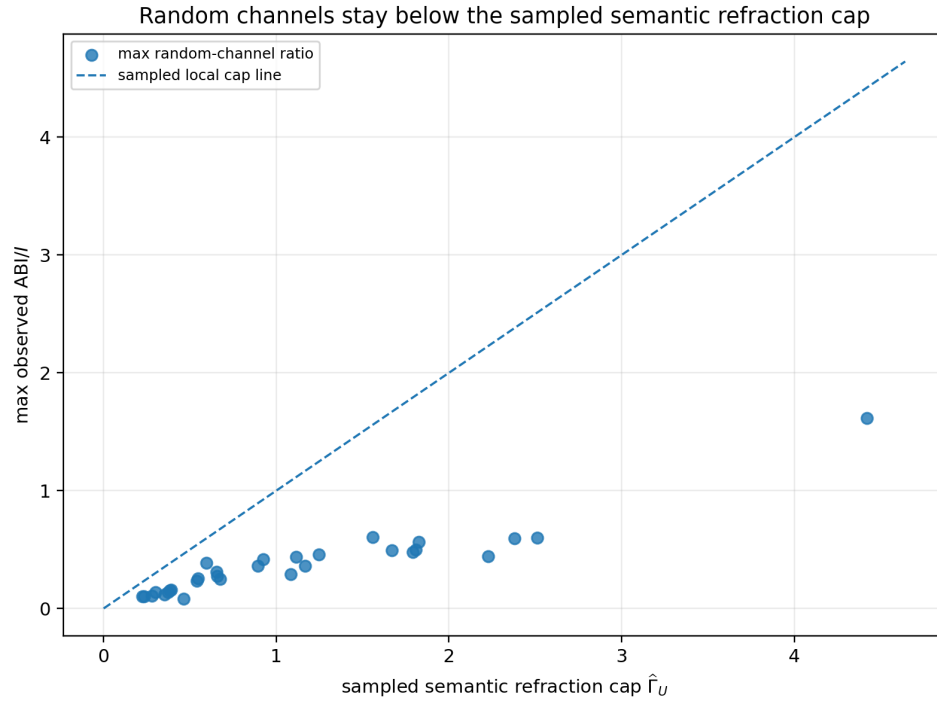


Figure 29: For random utilities and random channels, observed $\mathcal{I}/I$ ratios stayed below the sampled semantic refraction coefficient. Random channels were not optimized for local eigen-directions, so the ratios are typically far below the cap.

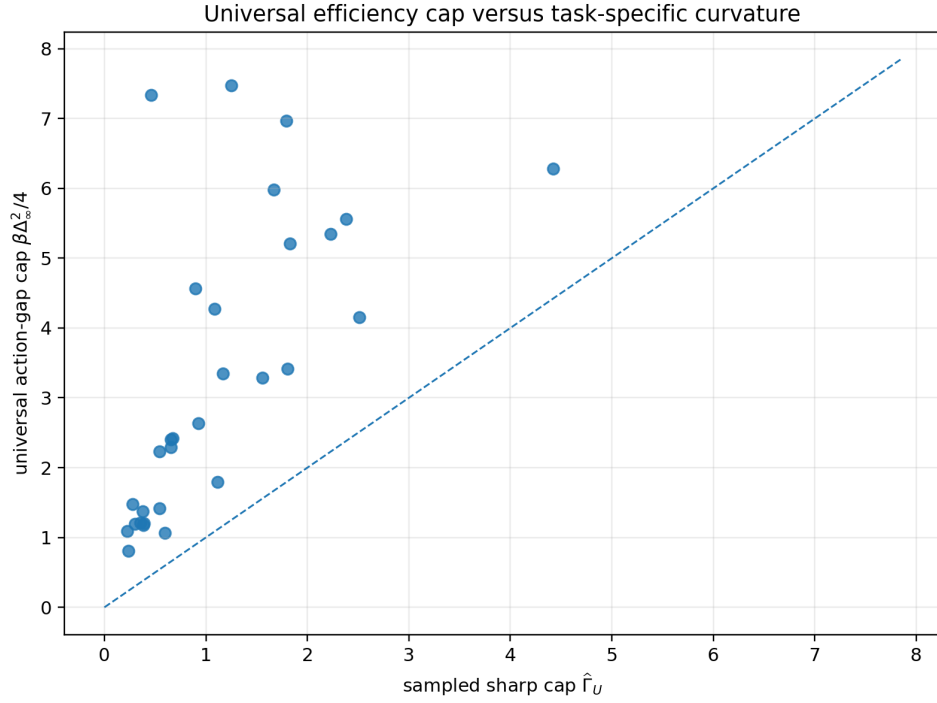


Figure 30: The universal action-gap cap is guaranteed but often loose compared with sampled task-specific curvature. This gap is a measure of utility geometry: not all action gaps align with reachable posterior directions.

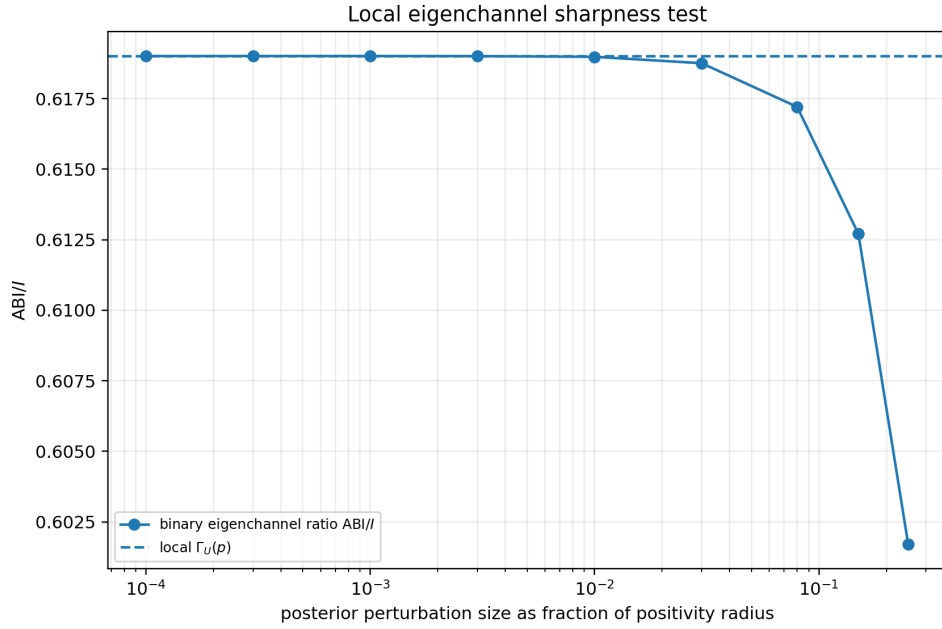


Figure 31: Local eigenchannel sharpness. Binary posterior perturbations in the top generalized-eigenvector direction converge to the predicted local refraction coefficient $\Gamma_{u,\beta}(p)$.

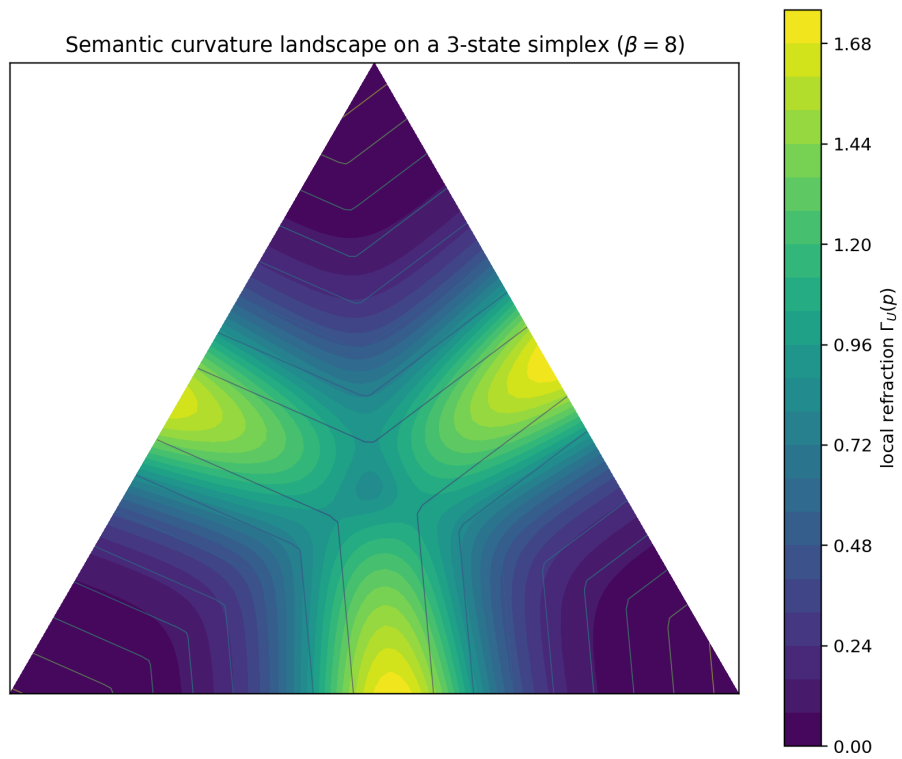


Figure 32: Semantic curvature landscape for a three-state, three-action utility. The conversion rate from Shannon nats to action value varies strongly across the belief simplex and concentrates near regions where the soft action policy is sensitive.

G.5 Interpretation

The central discovery is that useful information has a local refractive geometry. Shannon information supplies a Riemannian metric through negative entropy; the utility supplies another curvature through the action potential. The generalized eigenvalues of these two curvatures are the local exchange rates from nats to action value.

This reframes several earlier results:

- the semantic Fisher law is the low- β small-signal approximation of the same curvature ratio;
- the soft mist-flash cap is the binary special case of action-gap smoothness;
- semantic water-filling allocates information by following conductance curves, which are one-dimensional slices of the refraction geometry;
- dark information lives in zero-eigenvalue directions of the same geometry.

G.6 Applications and targets

- **Certified semantic compression.** If a representation has Shannon rate I , then ΓI is a certified upper bound on finite-rational action value. This can reject overclaims based on raw mutual information.
- **Task-aware channel design.** Channels should steer posterior perturbations toward high-refraction eigen-directions rather than simply maximizing total posterior movement.
- **Sensor auditing.** The action-gap cap gives a conservative certificate, while the sampled $\hat{\Gamma}$ gives a sharper empirical audit of useful leakage.
- **Privacy.** For a sensitive utility, $\Gamma_{\text{sensitive}} I$ bounds worst-case soft action leakage from a low-rate representation, while dark-capacity theory describes information that remains action-invisible.
- **Active learning.** Query policies can use local $\Gamma(p)$ to identify beliefs where one nat of evidence has the highest expected task value.

Further targets are:

1. compute $\Gamma_{u,\beta}$ exactly for multiclass top-1, top- r , abstention, and cost-sensitive classification;
2. develop convex or branch-and-bound algorithms for certifying the global supremum over the simplex;
3. combine semantic refraction with water-filling to solve multi-coordinate allocation under a global channel constraint;
4. derive finite-sample estimators for $\Gamma(p)$ from posterior samples and policy logits;
5. extend the theorem to continuous states by replacing the Fisher matrix with the information geometry of dominated densities.

H Threshold-abstention information: semantic sparks

Many decisions contain an explicit or implicit abstention threshold. A weak signal may reduce uncertainty, yet remain useless if it does not move the posterior far enough to justify action. This section studies the simplest such case and finds an exact value-information envelope. The result contrasts sharply with the binary mist-flash law without abstention: at very low information rates, diffuse weak evidence can be action-dormant, while rare threshold-crossing evidence can be optimal.

H.1 Binary abstention model

Let $Y \in \{0, 1\}$ with $P(Y = 1) = 1/2$. Write

$$q_s = P(Y = 1 \mid S = s), \quad T_s = \left| q_s - \frac{1}{2} \right|. \quad (200)$$

The Shannon mutual information is

$$I(Y; S) = \mathbb{E}\phi(T_s), \quad (201)$$

where

$$\phi(t) = \left(\frac{1}{2} + t \right) \log(1 + 2t) + \left(\frac{1}{2} - t \right) \log(1 - 2t), \quad 0 \leq t \leq \frac{1}{2}. \quad (202)$$

Suppose the decision maker may predict 0, predict 1, or abstain. After subtracting the common baseline $1/2$, let abstention have margin $h \in [0, 1/2]$. The hard value above abstention is then

$$\boxed{V_h(S) = \mathbb{E}(T_s - h)_+}. \quad (203)$$

The case $h = 0$ recovers ordinary binary prediction value $\mathbb{E}T_s$.

H.2 Exact spark envelope

Let $L = \log 2$. For $0 < h < 1/2$, define $t_*(h) \in (h, 1/2)$ as the solution of the tangent equation

$$\boxed{\phi(t_*) = (t_* - h)\phi'(t_*), \quad \phi'(t) = \log \frac{1 + 2t}{1 - 2t}}. \quad (204)$$

Equivalently, t_* maximizes the low-rate slope $(t - h)/\phi(t)$ over $t > h$.

Theorem 14 (Threshold-abstention value-information envelope). *For every binary channel with uniform input prior,*

$$\boxed{L_h(I(Y; S)) \leq V_h(S) \leq U_h(I(Y; S))}, \quad (205)$$

where

$$\boxed{U_h(I) = \begin{cases} \frac{t_* - h}{\phi(t_*)} I, & 0 \leq I \leq \phi(t_*), \\ \phi^{-1}(I) - h, & \phi(t_*) \leq I \leq L, \end{cases}} \quad (206)$$

and

$$\boxed{L_h(I) = \begin{cases} 0, & 0 \leq I \leq \phi(h), \\ \frac{1/2 - h}{L - \phi(h)} (I - \phi(h)), & \phi(h) \leq I \leq L. \end{cases}} \quad (207)$$

For $h = 0$, the upper envelope degenerates to $U_0(I) = \phi^{-1}(I)$ and the lower envelope is $L_0(I) = I/(2 \log 2)$.

Proof sketch. Any sign-symmetric distribution of posterior displacements $\pm T$ with $0 \leq T \leq 1/2$ is compatible with the uniform prior. Thus the optimization depends only on the distribution of T . Set $Z = \phi(T)$ and

$$v_h(Z) = (\phi^{-1}(Z) - h)_+. \quad (208)$$

The maximal expected value at fixed $\mathbb{E}Z = I$ is the upper concave envelope of v_h , while the minimal expected value is the lower convex envelope. Since ϕ is increasing and convex, ϕ^{-1} is concave. The upper concave envelope is therefore the tangent line from the origin to v_h , followed by v_h itself. The tangency condition is exactly $\phi(t_*) = (t_* - h)\phi'(t_*)$. The lower convex envelope is zero until $Z = \phi(h)$ and then the chord from $(\phi(h), 0)$ to $(\log 2, 1/2 - h)$. Symmetric posterior experiments with masses at 0, t_* , h , or $1/2$ attain the corresponding envelope points. $\square$

The upper-envelope construction is a *semantic spark*: below the tangent information $\phi(t_*)$, the optimal channel is mostly silent, but occasionally produces a posterior displacement of size t_* . The event is not a perfect revelation unless h is extremely high; it is tuned to just the right action-threshold overshoot.

For $h = 0.10$, the numerical constants are

$$\phi(h) = 0.0201355136, \quad t_* = 0.1946953417, \quad \phi(t_*) = 0.0778550284, \quad (209)$$

with low-rate spark slope

$$\frac{t_* - h}{\phi(t_*)} = 1.2163034763. \quad (210)$$

A rare perfect flash has slope $(1/2 - h)/\log 2 = 0.5770780164$. Thus the optimized spark converts low-rate Shannon information into thresholded action value more than twice as efficiently as rare perfect revelation in this example.

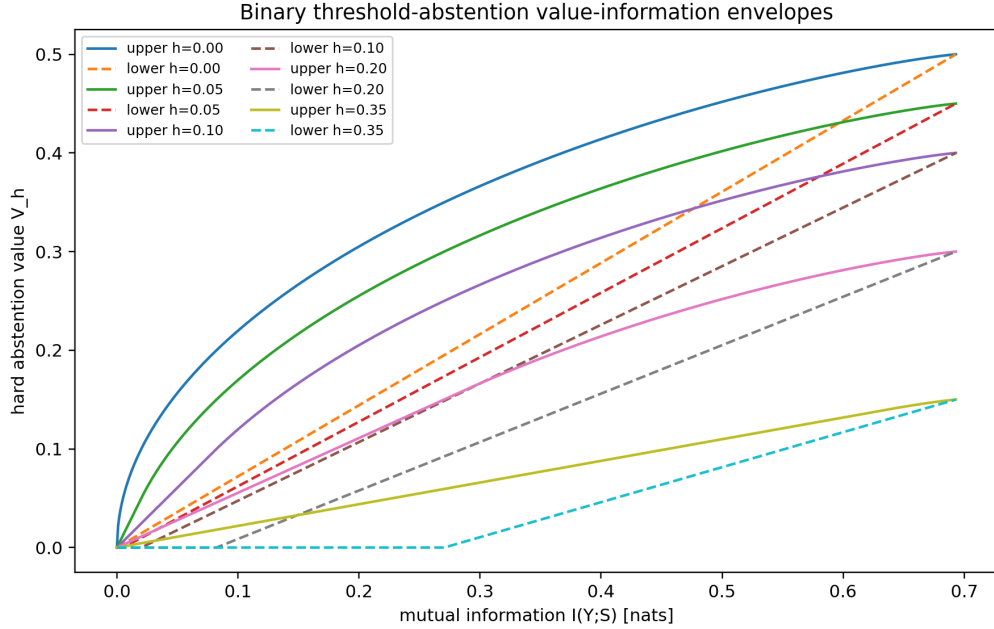


Figure 33: Exact value-information envelopes for binary hard-abstention tasks. The threshold creates a zero-value region in the lower envelope and a tangent-line spark branch in the upper envelope.

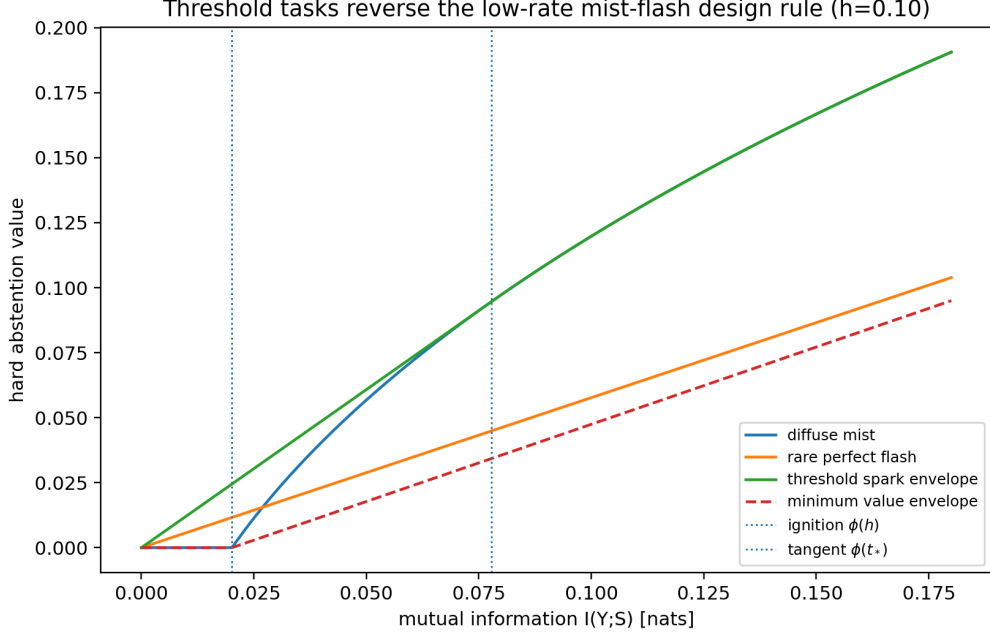


Figure 34: At $h = 0.10$, diffuse mist is action-dormant below $\phi(h)$, while a tuned rare spark has positive value for arbitrarily small mutual information. Thresholded tasks reverse the low-rate design preference of ordinary hard binary prediction.

H.3 Spark tangent laws

The tangent equation generates a one-parameter family of optimal spark amplitudes. The slope

$$s_h = \frac{t_*(h) - h}{\phi(t_*(h))} = \frac{1}{\phi'(t_*(h))} \quad (211)$$

is the best possible low-rate hard value per Shannon nat. The identity on the right follows directly from tangency. As h rises, $t_*(h)$ moves toward perfect revelation; as $h \downarrow 0$, the tangent collapses toward the origin and the square-root mist law of ordinary binary prediction reappears.

H.4 Soft abstention and dormant curvature

At finite rationality, the abstention threshold is softened rather than sharp. After subtracting the common $1/2$ baseline, let the three action utilities at posterior displacement $x = q - 1/2$ be

$$u_+(x) = x, \quad u_-(x) = -x, \quad u_\perp(x) = h. \quad (212)$$

The soft action potential is

$$F_{\beta,h}(x) = \frac{1}{\beta} \log \left(e^{\beta x} + e^{-\beta x} + e^{\beta h} \right). \quad (213)$$

At the prior $x = 0$,

$$F''_{\beta,h}(0) = \frac{2\beta}{2 + e^{\beta h}}. \quad (214)$$

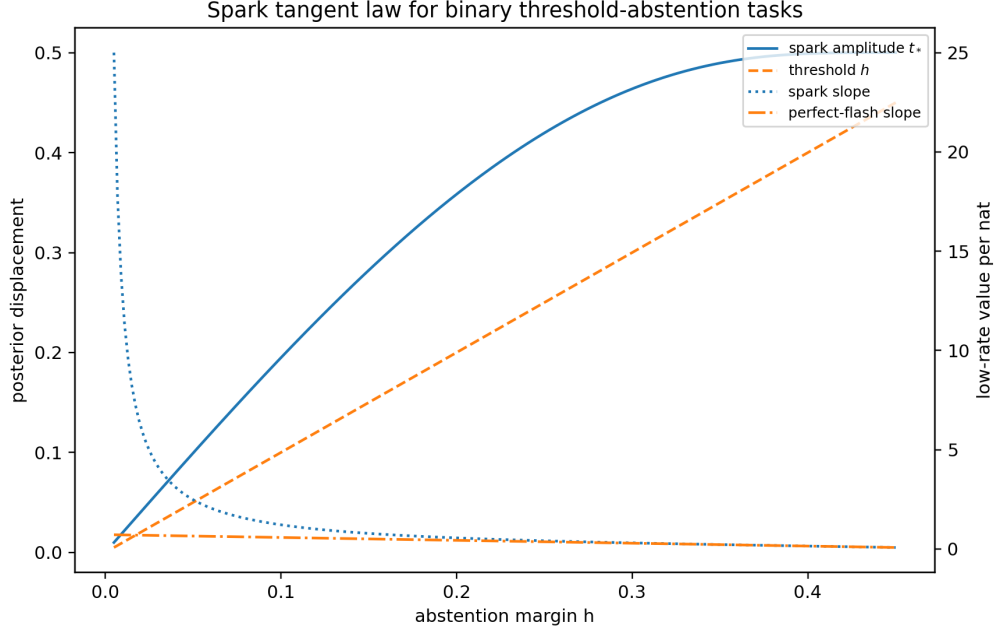


Figure 35: Spark tangent law. The optimal threshold-crossing posterior displacement $t_*(h)$ is above the abstention margin h . Its low-rate slope dominates the perfect-flash slope for moderate thresholds.

Consequently, for weak symmetric evidence around the prior, the local exchange rate from Shannon mutual information to soft ABI is

$$\Gamma_{\beta,h}^{(0)} = \frac{F''_{\beta,h}(0)}{4} = \frac{\beta}{2(2 + e^{\beta h})}. \quad (215)$$

This coefficient vanishes as $\beta \downarrow 0$ and also as $\beta \rightarrow \infty$ for fixed $h > 0$. The hard decision maker ignores subthreshold evidence, while an overly soft decision maker barely distinguishes the actions. The maximal prior sensitivity occurs at a universal scaled rationality.

Proposition 9 (Dormant-resonance constant). *For fixed abstention margin $h > 0$, $\Gamma_{\beta,h}^{(0)}$ is maximized when*

$$\boxed{\beta_* h = 1 + W(2/e) = 1.463055513365 \dots}, \quad (216)$$

where W is the Lambert W function. The maximal scaled exchange rate is

$$\boxed{h \max_{\beta} \Gamma_{\beta,h}^{(0)} = \frac{W(2/e)}{4} = 0.115763878341 \dots}. \quad (217)$$

Proof. Set $x = \beta h$. Up to the factor $1/h$, maximizing $\Gamma_{\beta,h}^{(0)}$ is equivalent to maximizing

$$\frac{x}{2(2 + e^x)}. \quad (218)$$

The first-order condition is $(x - 1)e^x = 2$, so $x = 1 + W(2/e)$. Substituting $e^x = 2/(x - 1)$ gives the stated maximum. $\square$

The soft curvature landscape also reveals where useful information becomes visible. For large β , $F''_{\beta,h}(x)$ is small near the prior, but concentrates near $x = \pm h$, the two hard action-switching surfaces.

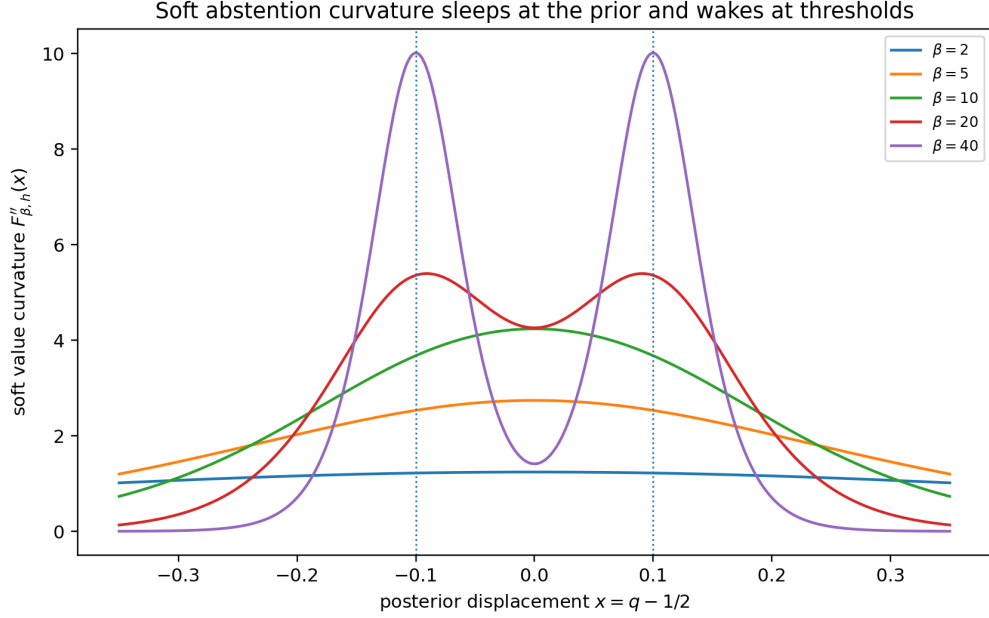


Figure 36: Soft abstention curvature for $h = 0.10$. As rationality increases, curvature at the prior becomes dormant and useful-information sensitivity migrates to the hard threshold surfaces $x = \pm h$.

H.5 Numerical stress tests

Random binary channels were sampled by drawing independent Dirichlet output laws for $P(S | Y = 0)$ and $P(S | Y = 1)$, using output alphabet sizes between 2 and 31. For each h , 3500 random channels were checked against the exact envelopes. No material violation was found.

h	channels	$\max(V_h - U_h)$	$\max(L_h - V_h)$
0.02	3500	-3.49×10^{-6}	0
0.05	3500	-8.24×10^{-7}	0
0.10	3500	-1.18×10^{-5}	0
0.20	3500	-5.84×10^{-7}	0
0.35	3500	-1.86×10^{-6}	0
0.45	3500	-1.38×10^{-6}	0

The zero entries in the lower-envelope residual column are caused by sampled channels whose evidence remains entirely below the abstention threshold, giving both $V_h = 0$ and $L_h = 0$ at their mutual information. The negative upper residuals indicate that all sampled values lay below the upper envelope, with the closest sampled channels within interpolation-scale margins.

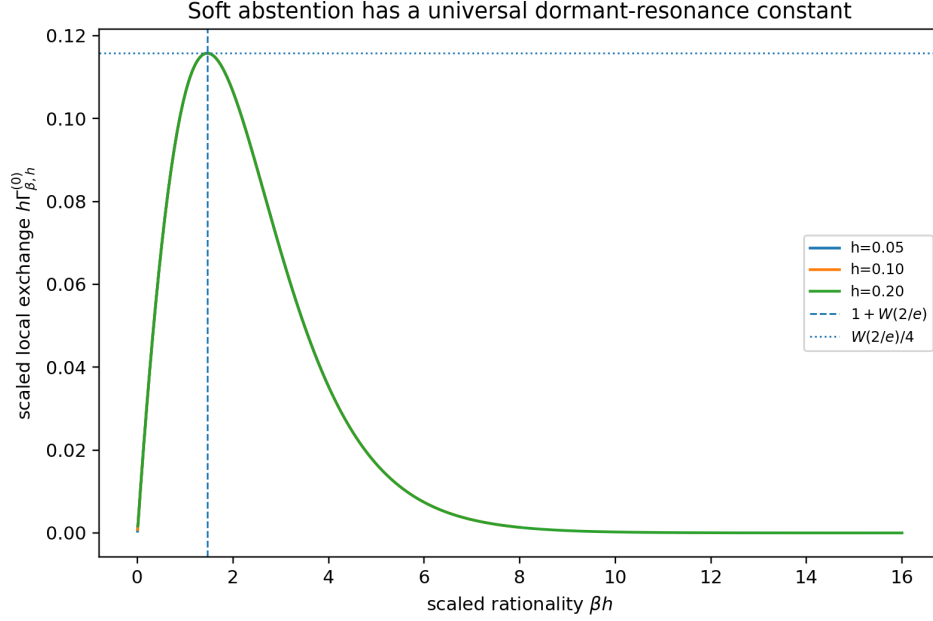


Figure 37: Universal dormant-resonance collapse. The scaled local exchange rate $h\Gamma_{\beta,h}^{(0)}$ peaks at $\beta h = 1 + W(2/e)$ for all tested thresholds.

H.6 Interpretation and applications

The central discovery is that an action threshold changes the low-rate geometry of useful information:

Without a threshold: low-rate mist can be optimal. With a threshold: calibrated rare sparks can be optimal.

(219)

This does not contradict the earlier mist–flash law. It identifies a different decision geometry: the hard value function contains a flat region. Shannon information can accumulate inside that flat region without producing hard action value.

Potential applications include:

- **Selective classification and abstaining classifiers.** Training signals should be evaluated by how often they cross useful decision thresholds, not only by average mutual information with the label.
- **Medical and safety triage.** Low-rate monitoring may be best designed as threshold sparks when action is only warranted above a risk margin.
- **Semantic communication.** A communication link for alarms should not necessarily maximize weak reconstruction quality; it may need to allocate bits to calibrated threshold crossings.
- **Privacy auditing.** A representation with small mutual information can still have high thresholded action value if its information is concentrated into sparks.
- **Human-AI decision support.** Soft-abstention resonance predicts an intermediate rationality or confidence-temperature at which subthreshold evidence is most policy-visible.

Further targets are:

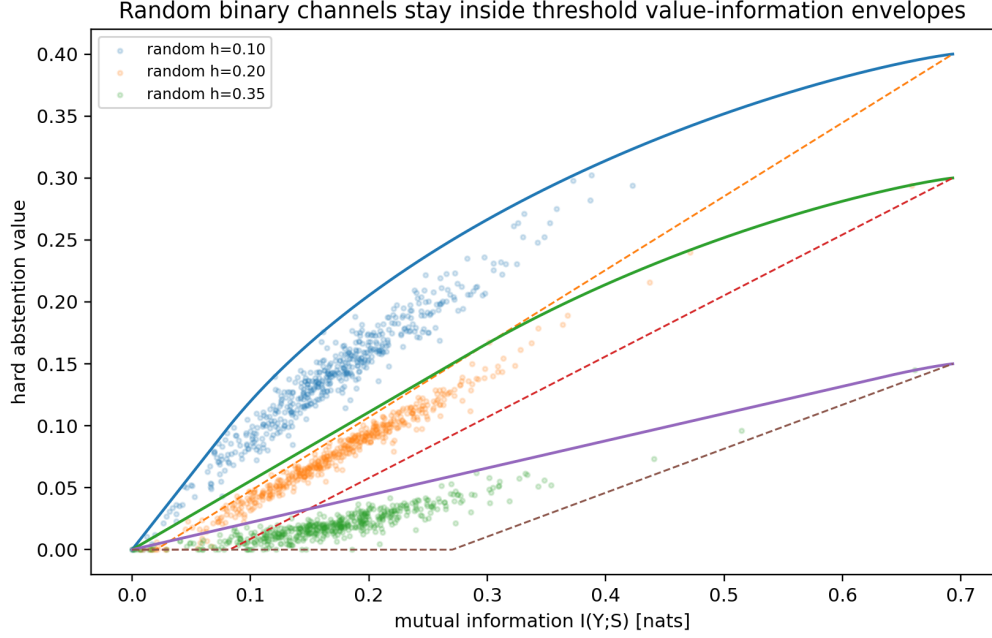


Figure 38: Random-channel stress test. Sampled channels lie between the lower and upper hard-abstention envelopes for representative thresholds.

1. prove multiclass analogues for abstention and top- r selective classification;
2. derive finite- β value-information envelopes beyond the local dormant-curvature law;
3. extend spark envelopes to unequal priors and asymmetric false-positive/false-negative costs;
4. combine semantic water-filling with threshold sparks across multiple coordinates;
5. study sequential spark accumulation, where individually subthreshold messages may cross the threshold after causal aggregation;
6. build estimators that separate raw mutual information, dormant information, and spark information in empirical classifiers.

I Rationality spectra and ghost extinction

The previous sections treat the inverse rationality parameter β as a modeling knob. The next observation is that β is itself an information-theoretic coordinate. A signal can be valuable for a moderately stochastic decision maker and nearly valueless for a perfectly hard decision maker, even when the utility function and the statistical experiment are fixed. Thus the map

$$\boxed{\beta \mapsto \mathcal{I}_{u,\beta}(Y; S)} \quad (220)$$

will be called the *rationality spectrum* of the signal.

This section records a new exact slope identity, a non-monotonicity mechanism, and two high-rationality extinction laws. The exceptional point is that increasing rationality can erase confidence-shaping information. In this sense semantic information has a band-pass component: some evidence is visible only to agents with a finite softness of action.

I.1 Rationality-slope identity

Let

$$Z_\beta(p) = \sum_a \pi(a) e^{\beta \langle u_a, p \rangle}, \quad F_\beta(p) = \beta^{-1} \log Z_\beta(p), \quad (221)$$

and let

$$\omega_{\beta,p}(a) = \frac{\pi(a) e^{\beta \langle u_a, p \rangle}}{Z_\beta(p)} \quad (222)$$

be the soft action policy induced by belief p . Define the action-policy concentration

$$K_\beta(p) = D_{\text{KL}}(\omega_{\beta,p} \| \pi). \quad (223)$$

Theorem 15 (Rationality-slope identity). *For any finite decision problem and signal S ,*

$$\boxed{\frac{d}{d\beta} \mathcal{I}_{u,\beta}(Y; S) = \frac{1}{\beta^2} [\mathbb{E}_S K_\beta(P_{Y|S}) - K_\beta(P_Y)]}. \quad (224)$$

Equivalently, the sign of the rationality response is the sign of a Jensen gap of action-policy concentration, not the sign of a Shannon information quantity.

Proof. Let $v_a(p) = \langle u_a, p \rangle$. Since

$$\partial_\beta \log Z_\beta(p) = \sum_a \omega_{\beta,p}(a) v_a(p), \quad (225)$$

we have

$$\frac{\partial}{\partial \beta} F_\beta(p) = \frac{1}{\beta} \sum_a \omega_{\beta,p}(a) v_a(p) - \frac{1}{\beta^2} \log Z_\beta(p) \quad (226)$$

$$= \frac{1}{\beta^2} \sum_a \omega_{\beta,p}(a) [\log \omega_{\beta,p}(a) - \log \pi(a)] \quad (227)$$

$$= \frac{K_\beta(p)}{\beta^2}. \quad (228)$$

Taking the posterior Jensen gap gives the result. $\square$

There is also a heat-capacity form. Differentiating $K_\beta(p) = \beta \partial_\beta \log Z_\beta(p) - \log Z_\beta(p)$ gives

$$\frac{d}{d\beta} K_\beta(p) = \beta \text{Var}_{a \sim \omega_{\beta,p}}(v_a(p)). \quad (229)$$

Consequently

$$\boxed{\frac{d}{d\beta} \left(\beta^2 \frac{d}{d\beta} \mathcal{I}_{u,\beta}(Y; S) \right) = \beta \left[\mathbb{E}_S \text{Var}_{\omega_{\beta, P_Y|S}}(v_a(P_{Y|S})) - \text{Var}_{\omega_{\beta, P_Y}}(v_a(P_Y)) \right]}. \quad (230)$$

This identifies rationality curvature with a posterior gap in soft action-score variance.

I.2 Non-monotonicity and semantic band-pass behavior

The slope identity shows why no monotonicity law in β should be expected. A posterior experiment can increase action-policy concentration at small β and decrease the concentration gap at large β .

The simplest exact mechanism is binary prediction with a prior margin $m > 0$. Let a signal move the posterior margin to

$$\delta_S = m + X_S, \quad \mathbb{E}X_S = 0, \quad \delta_S > 0 \quad \text{a.s.} \quad (231)$$

The hard Bayes action never changes, so

$$\mathcal{I}_\infty(Y; S) = \mathbb{E}|\delta_S| - |m| = 0. \quad (232)$$

For finite rationality,

$$\boxed{\mathcal{I}_\beta(Y; S) = \frac{1}{\beta} [\mathbb{E} \log \cosh(\beta \delta_S) - \log \cosh(\beta m)]}. \quad (233)$$

For the symmetric two-point experiment $\delta_S = m \pm a$ with equal probabilities and $0 < a < m$,

$$\mathcal{I}_\beta = \frac{1}{2\beta} [\log \cosh \beta(m - a) + \log \cosh \beta(m + a)] - \frac{1}{\beta} \log \cosh \beta m. \quad (234)$$

At small β ,

$$\mathcal{I}_\beta = \frac{\beta a^2}{2} + O(\beta^3) > 0, \quad (235)$$

while $\mathcal{I}_\beta \rightarrow 0$ as $\beta \rightarrow \infty$. Therefore the rationality spectrum is non-monotone. This produces a clean separation between *choice-changing* information and *confidence-shaping ghost information*.

In a random stress test with 450 finite decision channels with $K = 4$ world states, 3 actions, and 5 posterior atoms, 44.89% of analyzed examples had a terminal drop of at least 8% from the maximum ABI on the tested range. This is not a universal frequency claim; it is evidence that non-monotonicity is a common finite-dimensional phenomenon rather than a carefully engineered pathology.

I.3 Derivative verification

The rationality-slope identity was checked by finite differences on random finite decision problems. The maximum absolute residual across 1200 derivative checks was

$$2.68 \times 10^{-10}, \quad (236)$$

with median absolute residual

$$8.49 \times 10^{-12}. \quad (237)$$

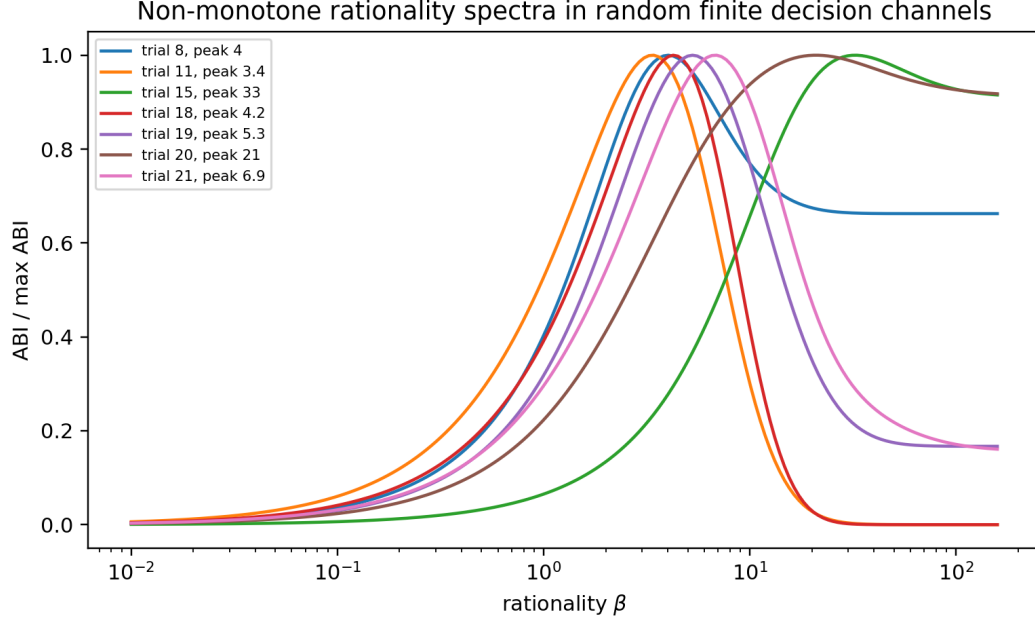


Figure 39: Random finite decision channels often have non-monotone rationality spectra. Each curve is normalized by its maximum ABI. The existence of interior peaks means that increasing rationality can reduce the semantic value of a fixed signal.

I.4 Finite-atom ghost extinction law

The high-rationality decay of ghost information is controlled by the closest posterior approach to a decision boundary. In binary prediction, assume $\delta_S > 0$ almost surely, let

$$\delta_{\min} = \min_s \delta_s, \quad (238)$$

and let $w_{\min}$ be the total probability of posterior atoms with margin $\delta_{\min}$. If $\delta_{\min} < m$, then

$$\boxed{\mathcal{I}_\beta(Y; S) = \frac{w_{\min}}{\beta} e^{-2\beta\delta_{\min}} (1 + o(1)).} \quad (239)$$

Thus a hard decision maker sees zero value, while a finite-rationality decision maker sees a ghost value that decays exponentially at a rate set by the nearest posterior margin.

For the tested case $m = 0.25$, $a = 0.20$, the nearest margin is $\delta_{\min} = 0.05$ and $w_{\min} = 1/2$. The scaled tail

$$\beta e^{2\beta\delta_{\min}} \mathcal{I}_\beta \quad (240)$$

converged numerically to 0.4999997888, matching the predicted constant $1/2$.

I.5 Boundary-density ghost extinction law

The finite-atom law changes when posterior margins can approach the boundary continuously. Suppose $\delta_S > 0$ almost surely, $\mathbb{E}\delta_S = m$, and the density of δ_S near zero has the form

$$f(\delta) = c\delta^\alpha + o(\delta^\alpha), \quad \alpha > -1. \quad (241)$$

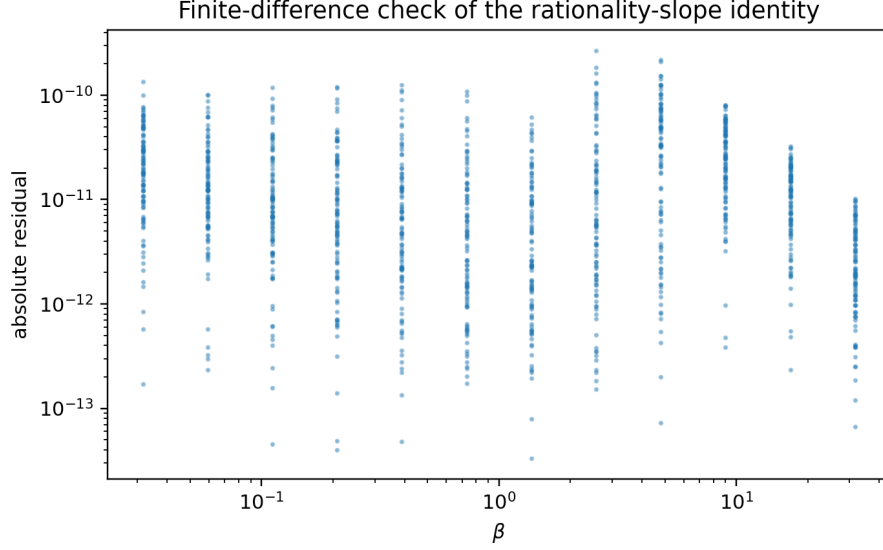


Figure 40: Finite-difference residuals for the rationality-slope identity. The residuals remain at numerical differentiation scale over the tested range of β .

Then

$$\mathcal{I}_\beta(Y; S) \sim \frac{c \Gamma(\alpha + 1) \eta_D(\alpha + 2)}{2^{\alpha+1}} \beta^{-(\alpha+2)}, \quad \beta \rightarrow \infty, \quad (242)$$

where

$$\eta_D(s) = \sum_{n \geq 1} (-1)^{n+1} n^{-s} = (1 - 2^{1-s}) \zeta(s) \quad (243)$$

is the Dirichlet eta function.

For a uniform margin distribution on $[0, 2m]$, $c = f(0) = 1/(2m)$ and $\alpha = 0$, so

$$\mathcal{I}_\beta(Y; S) \sim \frac{\pi^2}{48m} \beta^{-2}. \quad (244)$$

At $m = 0.25$, the predicted constant is

$$\frac{\pi^2}{48m} = 0.822467033424113 \dots \quad (245)$$

The numerical experiment gave

$$\beta^2 \mathcal{I}_\beta = 0.822467033424645 \quad (246)$$

at the largest tested β .

I.6 Rationality peak map

For two-point choice-preserving evidence $\delta = m \pm a$, the location and height of the rationality peak depend on the amplitude ratio a/m . As a/m increases, the closest posterior margin $m - a$ approaches the decision boundary, and the optimal rationality shifts accordingly.

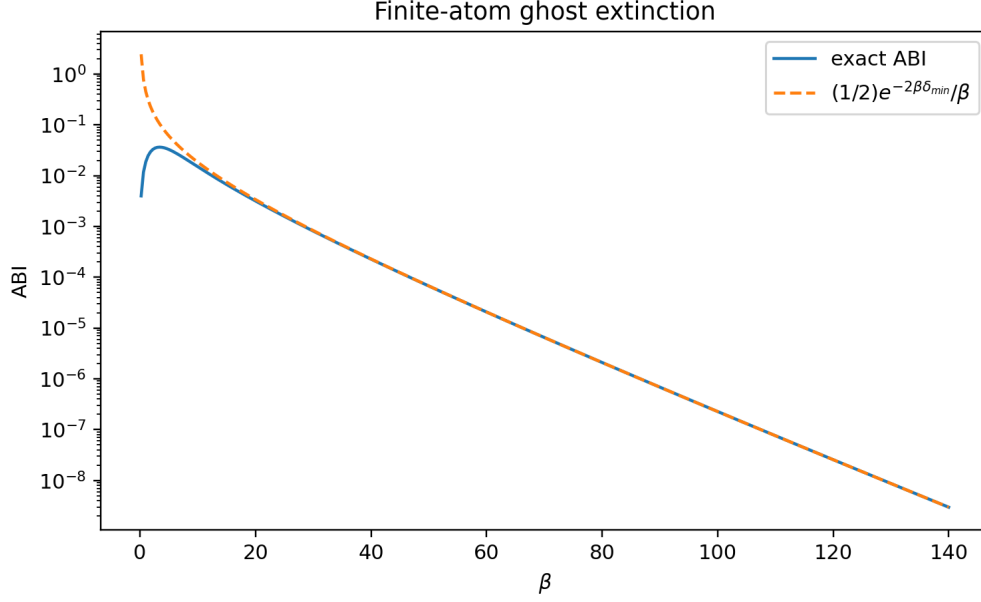


Figure 41: Finite-atom ghost extinction. The finite-rationality value of choice-preserving evidence dies as $(1/2)e^{-2\beta\delta_{\min}}/\beta$ in the tested two-point case.

The largest peak height in the tested grid was

$$\max_{a/m} \frac{\sup_{\beta} \mathcal{I}_{\beta}}{m} \approx 0.20705. \quad (247)$$

Thus confidence-shaping information can have substantial finite-rational value even though its hard-rational value is exactly zero.

I.7 Multi-action boundary conjecture

The binary laws suggest a general action-boundary principle. Let $a^*(p)$ be the unique optimal action at belief p , and let

$$g_b(p) = \langle u_{a^*(p)} - u_b, p \rangle, \quad b \neq a^*(p), \quad (248)$$

be the action gaps. If a signal is choice-preserving in the sense that $a^*(P_{Y|S}) = a^*(P_Y)$ almost surely, then

$$\mathcal{I}_{u,\infty}(Y; S) = 0. \quad (249)$$

The following conjectural asymptotic generalizes the binary ghost law:

$$\boxed{\mathcal{I}_{u,\beta}(Y; S) \approx \frac{1}{\beta} \left[\mathbb{E}_S \sum_{b \neq a^*} e^{-\beta g_b(P_{Y|S})} - \sum_{b \neq a^*} e^{-\beta g_b(P_Y)} \right]}. \quad (250)$$

Consequently, the high-rationality tail is controlled by the posterior distribution of distances to action-indifference hyperplanes. This would turn rationality spectra into a kind of Laplace spectroscopy for semantic margins.

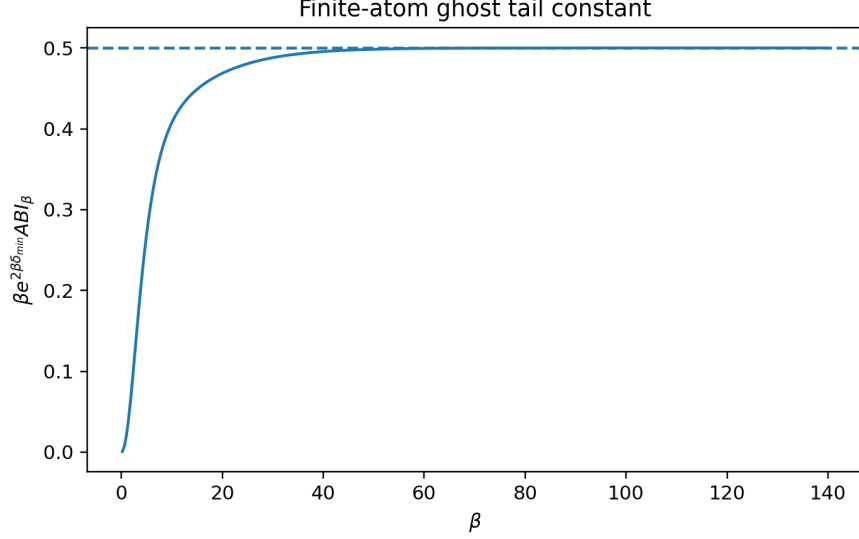


Figure 42: Scaled finite-atom tail constant. The numerical tail converges to $w_{\min} = 1/2$.

I.8 New quantities suggested by the spectrum

The rationality spectrum motivates several task-relative diagnostics:

$$R_{\infty}(S) = \lim_{\beta \rightarrow \infty} \mathcal{I}_{u,\beta}(Y; S), \quad \text{hard choice-changing residue,} \quad (251)$$

$$G_*(S) = \sup_{\beta > 0} \mathcal{I}_{u,\beta}(Y; S) - R_{\infty}(S), \quad \text{ghost surplus,} \quad (252)$$

$$\beta_*(S) = \arg \max_{\beta > 0} \mathcal{I}_{u,\beta}(Y; S), \quad \text{semantic resonance rationality,} \quad (253)$$

$$\Lambda(S) = \limsup_{\beta \rightarrow \infty} -\frac{1}{\beta} \log(\beta |\mathcal{I}_{u,\beta} - R_{\infty}|), \quad \text{boundary extinction rate.} \quad (254)$$

For binary finite-atom ghost evidence, $\Lambda = 2\delta_{\min}$. These quantities are invisible to Shannon mutual information because they depend on how posterior movement sits relative to action boundaries.

I.9 Applications

Semantic communication. A communication system optimized for a hard downstream classifier may discard confidence-shaping evidence. A finite-rational receiver, a human operator, or a soft controller may value exactly that evidence. Rationality spectra provide a way to specify the receiver class rather than optimizing for a single value of β .

Safety alarms and triage. Weak evidence that never changes the top action can still alter urgency, confidence, or escalation policy. The ghost extinction laws quantify how such evidence disappears as the policy becomes deterministic.

Privacy. A signal can have zero hard action value but positive finite-rational policy value. Privacy audits based only on hard decisions may miss confidence leakage. Conversely, spectra can distinguish durable choice-changing leakage from ghost leakage that depends on a receiver’s softness.

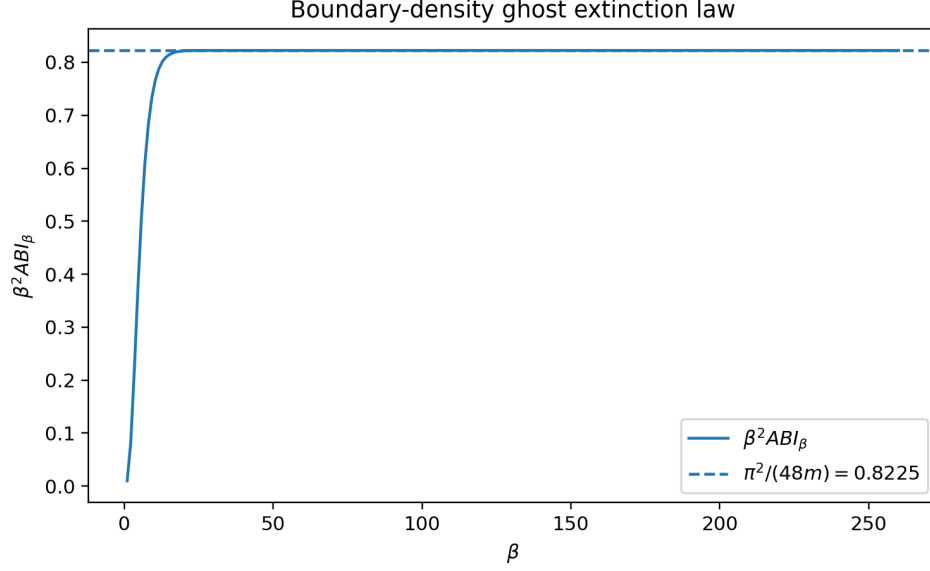


Figure 43: Boundary-density ghost extinction. When the posterior margin distribution touches the action boundary with nonzero density, ghost information dies polynomially rather than exponentially.

Human and model calibration. The resonance rationality β_* identifies the decision softness at which a signal is maximally useful. This may be measurable experimentally by varying response stochasticity, temperature, or confidence thresholds.

Representation learning. A representation can be scored by its entire rationality spectrum. Two representations with identical hard accuracy can differ sharply in ghost surplus, boundary extinction rate, and finite-rational robustness.

I.10 Targets for further study

1. Prove the multi-action boundary conjecture under explicit action-gap regularity conditions.
2. Classify when rationality spectra are unimodal, multimodal, or monotone.
3. Develop inverse methods that recover semantic margin distributions from the high- β tail of $\mathcal{I}_{u,\beta}$.
4. Extend the finite-atom and boundary-density laws to abstention, rejection, and top- k decision utilities.
5. Define a rationality-spectrum rate-distortion theory in which encoders preserve $\mathcal{I}_{u,\beta}$ over a range of β values rather than at a single rationality.
6. Study whether neural-network temperature scaling can be interpreted as moving along an empirical ABI rationality spectrum.

I.11 Data added in v1.0

The v1.0 experiments added the following reproducibility artifacts:

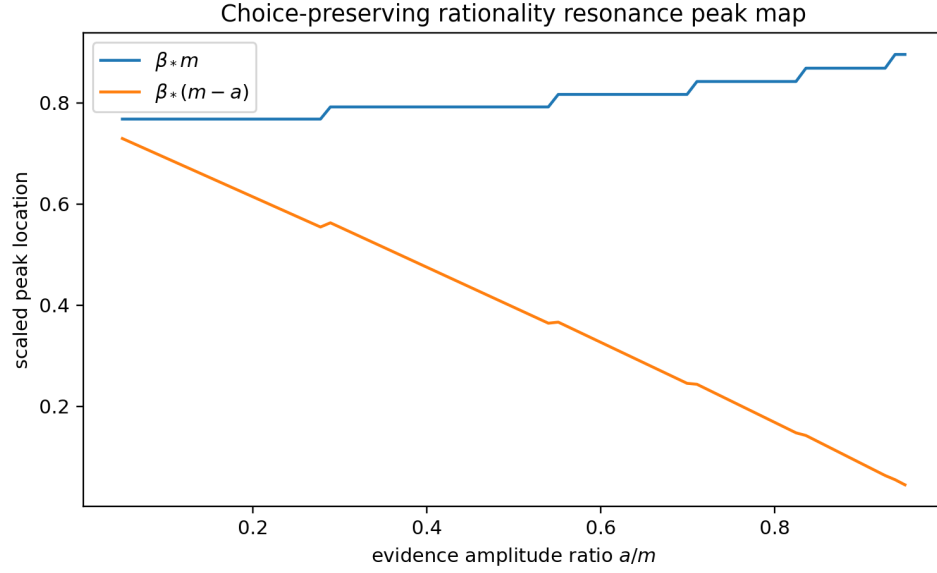


Figure 44: Peak map for two-point choice-preserving evidence. The resonant rationality is controlled jointly by the prior margin m and the nearest posterior margin $m - a$.

- `scripts/experiments_v1_0.py`: rationality-spectrum, slope-identity, and ghost-extinction experiments;
- `data/rationality_spectra_random_summary_v1_0.csv`: random finite-channel rationality spectra;
- `data/rationality_slope_identity_residuals_v1_0.csv`: finite-difference checks of the slope theorem;
- `data/binary_ghost_finite_atom_extinction_v1_0.csv`: finite-atom ghost tail data;
- `data/binary_ghost_boundary_density_v1_0.csv`: boundary-density ghost tail data;
- `data/binary_ghost_peak_phase_map_v1_0.csv`: two-point resonance peak map;
- `logs/research_log_v1_0.md`: research log for this round.

J Experimental reproducibility

The source bundle accompanying this draft contains:

- `scripts/experiments_v0_2.py`: v0.2 experiment and figure-generation script;
- `scripts/experiments_v0_3.py`: semantic eigenlens and compression experiments;
- `scripts/experiments_v0_4.py`: action-policy KL and rationality-resonance experiments;
- `data/random_channel_envelope_stress.csv`: random-channel residuals;
- `data/small_information_value_ratios.csv`: small-information ratios;
- `data/crystal_ladder_segments.csv`: exact lower-envelope segment data;
- `data/semantic_eigenlens_random_perturbations_v0_3.csv`: random local-channel eigenlens tests;
- `data/semantic_compression_tradeoff_v0_3.csv`: posterior compression experiments;
- `data/policy_kl_identity_residuals_v0_4.csv`: random-channel checks of the action-policy KL identity;
- `data/choice_preserving_resonance_curves_v0_4.csv`: soft ABI curves for choice-preserving evidence;
- `data/resonance_peak_drift_v0_4.csv`: finite-amplitude drift of the resonance peak;
- `figures/*.png`: rendered experimental figures;
- `logs/research_log_v0_2.md`: v0.2 research log;
- `logs/research_log_v0_3.md`: v0.3 research log;
- `logs/v0_4_experiment_log.txt`: v0.4 experiment log;
- `scripts/experiments_v0_8.py`: semantic water-filling allocation and threshold experiments;
- `data/semantic_water_filling_allocations_v0_8.csv`: optimal allocation curves;
- `data/semantic_activation_thresholds_v0_8.csv`: finite-rational activation budgets;
- `scripts/experiments_v0_9.py`: semantic refraction and relative-smoothness experiments;
- `data/relative_smoothness_random_channel_stress_v0_9.csv`: random-channel cap stress tests;
- `data/local_eigenchannel_sharpness_v0_9.csv`: local eigenchannel convergence data;
- `logs/research_log_v0_8.md`: v0.8 research log;
- `logs/research_log_v0_9.md`: v0.9 research log;
- `scripts/experiments_v0_10.py`: threshold-abstention envelope and soft-abstention experiments;
- `data/threshold_envelope_random_channel_stress_v0_10.csv`: random-channel checks of threshold envelopes;
- `data/threshold_tangent_spark_laws_v0_10.csv`: tangent amplitudes, ignition information, and low-rate spark slopes;

- `data/soft_abstention_dormant_resonance_v0.10.csv`: universal dormant-resonance checks;
- `logs/research_log_v0.10.md`: v0.10 research log;
- `scripts/experiments_v1.0.py`: rationality-spectrum, slope-identity, and ghost-extinction experiments;
- `data/rationality_spectra_random_summary_v1.0.csv`: random finite-channel rationality spectra;
- `data/rationality_slope_identity_residuals_v1.0.csv`: finite-difference checks of the rationality-slope theorem;
- `data/binary_ghost_finite_atom_extinction_v1.0.csv`: finite-atom ghost tail data;
- `data/binary_ghost_boundary_density_v1.0.csv`: boundary-density ghost tail data;
- `logs/research_log_v1.0.md`: v1.0 research log;

References

- [1] C. E. Shannon. A mathematical theory of communication. *Bell System Technical Journal*, 27:379–423 and 623–656, 1948.
- [2] D. Blackwell. Comparison of experiments. In *Proceedings of the Second Berkeley Symposium on Mathematical Statistics and Probability*, pages 93–102, 1951.
- [3] D. Blackwell. Equivalent comparisons of experiments. *Annals of Mathematical Statistics*, 24(2):265–272, 1953.
- [4] H. de Oliveira. Blackwell’s informativeness theorem using diagrams. *Games and Economic Behavior*, 109:126–131, 2018.
- [5] A. Banerjee, S. Merugu, I. S. Dhillon, and J. Ghosh. Clustering with Bregman divergences. *Journal of Machine Learning Research*, 6:1705–1749, 2005.
- [6] P. S. Chodrow. Equivalence of informations characterizes Bregman divergences. *Entropy*, 27(7):766, 2025.
- [7] J. Massey. Causality, feedback and directed information. In *Proceedings of the International Symposium on Information Theory and its Applications*, 1990.
- [8] D. Tsur. Directed information. arXiv:2602.09711, 2026.
- [9] J. Scarlett and V. Cevher. An introductory guide to Fano’s inequality with applications in statistical estimation. arXiv:1901.00555, 2019.
- [10] Survey authors. Theory of Mind, Generative AI, and Deep Joint Source-Channel Coding for semantic communication. arXiv:2502.16468, 2025.
- [11] Survey authors. A survey on semantic communication for vision. arXiv:2601.22202, 2026.
- [12] A. Kamatsuka and T. Yoshida. A generalized information bottleneck method: A decision-theoretic perspective. arXiv:2602.18405, 2026.

- [13] Y. Zhou, C. Peng, R. Wang, Y. Xiao, Y. Li, G. Shi, and P. Zhang. Distributed information bottleneck theory for multi-modal task-aware semantic communication. arXiv:2510.04000, 2025.
- [14] C. P. Chambers, Y. Masatlioglu, P. Natenzon, and C. Raymond. Measuring choice difficulty. arXiv:2604.26761, 2026.
- [15] A. Xu. One if by land, two if by sea, three if by four seas, and more to come: Values of perception, prediction, communication, and common sense in decision making. arXiv:2601.06077, 2026.